Theories as Regularizers*

Amedeo Andriollo Juan F. Imbet

March 16, 2026

Abstract

Machine-learning methods for return prediction ignore economic theory. Structural asset-pricing models encode deep knowledge about expected returns but are individually misspecified. We let the data decide which theories help and how much. We embed over fifty model-implied restrictions—from consumption-based, production-based, intermediary, behavioral, and demand-system frameworks—as soft penalties in a kernel ridge regression estimator. Each restriction carries its own penalty weight, selected by cross-validation: theories that hurt prediction are silenced; theories that help are amplified. The result is the first empirical ranking of economic theories by their marginal out-of-sample forecasting value. Theory-informed penalization delivers higher out-of-sample predictive accuracy and Sharpe ratios than both agnostic machine learning and standard kernel methods. Consumption-based and intermediary restrictions contribute the most. Theory helps disproportionately when data are scarce and during crises—precisely when investors need guidance most.

JEL Classification: G12, G17, C14, C52.
Keywords: Asset pricing, kernel ridge regression, structural restrictions, machine learning, model selection.

*We thank [acknowledgments]. Andriollo: Paris Dauphine University – PSL, amedeo.andriollo@dauphine.psl.eu. Imbet: Paris Dauphine University – PSL, juan.imbet@dauphine.psl.eu.

1 Introduction

Two traditions compete for the task of predicting asset returns. Statistical machine-learning methods—neural networks, random forests, gradient-boosted trees—estimate flexible functions of observable characteristics and deliver strong out-of-sample (OOS) performance, yet they treat economic theory as irrelevant input (Gu et al., 2020). Structural asset-pricing models—CAPM, long-run risk, habit formation, intermediary-based pricing—derive sharp restrictions on the pricing kernel and expected returns, yet each model is misspecified in isolation. One tradition has power without guidance; the other has guidance without robustness. Neither alone is satisfactory.

We propose a framework that draws on both. Our estimator, Theory-Informed Kernel Ridge Regression (Theory-KRR), embeds fifty-six model-implied structural restrictions as soft penalties in the KRR objective. Each restriction j enters with its own penalty multiplier μ_j, and the entire vector $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_J)$ is selected by cross-validation. The mechanism is automatic: restrictions from misspecified theories receive $\mu_j \to 0$—the data shut them down—while restrictions from empirically useful theories receive $\mu_j > 0$, steering the estimator toward economically sensible regions of the reproducing kernel Hilbert space $\mathcal{H}$. No single structural model needs to be correct. The data adjudicate.

The restrictions we impose are not generic shape constraints such as monotonicity or convexity (Bryzgalova et al., 2023b). They are genuine model-implied restrictions: Euler-equation moment conditions, factor-loading constraints, demand-system equilibrium conditions, and stochastic discount factor bounds, each derived from a specific economic theory. By cataloging fifty-six such restrictions across consumption-based, production-based, intermediary, behavioral, and demand-system frameworks, we construct a comprehensive menu of theoretical guidance. Cross-validation then selects which theories—and which specific restrictions within each theory—contribute to OOS prediction.

A methodological parallel clarifies the logic. The LLM-Lasso of Bybee et al. (2025) replaces the agnostic ℓ_1 penalty in Lasso with heterogeneous, text-informed penalties: large-

language-model predictions set a prior over which covariates matter, and cross-validation adjusts a single scalar that scales the text-derived penalty vector. Our approach applies the same principle in a different domain. Where LLM-Lasso uses text priors to inform penalization in *parameter space*, Theory-KRR uses economic-theory priors to inform penalization in *function space*. The mathematical structure is analogous: both replace a single, uninformed regularization parameter with a vector of informed, heterogeneous penalties, and both let cross-validation arbitrate between the prior and the data. Theory-KRR is, in this sense, the function-space generalization of LLM-Lasso.

What does this framework deliver? Theory-KRR outperforms standard KRR and several benchmark ML methods in OOS return prediction, measured by R^2_{OOS} and portfolio Sharpe ratios. The improvement is not uniform: theory-informed penalization helps most in small samples and during financial crises, precisely when purely statistical methods overfit and economic structure is most valuable. Beyond predictive gains, the cross-validated multipliers $\boldsymbol{\mu}$ produce an empirical ranking of economic theories by their marginal OOS forecasting value. Consumption-based restrictions (Campbell and Cochrane, 1999; Bansal and Yaron, 2004) and intermediary-based restrictions (He and Krishnamurthy, 2013; Adrian et al., 2014) receive the largest multipliers, indicating that these theories provide the most useful guidance for return prediction. Production-based and behavioral restrictions contribute modestly. Several models receive μ_j indistinguishable from zero, consistent with those theories being empirically irrelevant for forecasting in our sample.

This paper makes three contributions. First, we develop a unified framework that embeds fifty-six structural restrictions from diverse asset-pricing models as soft penalties in a single KRR estimator. The framework nests standard KRR (all $\mu_j = 0$) and hard-constrained estimation (any $\mu_j \to \infty$) as special cases, and it accommodates restrictions of heterogeneous functional form—moment conditions, inequality constraints, and functional equations—within a single objective. Second, we establish a formal connection between theory-informed function-space penalization and the LLM-Lasso approach to parameter-

space penalization. This connection unifies two strands of the literature—structural econometrics and text-based ML—under a common principle of informed regularization. Third, we produce the first empirical ranking of economic theories by their marginal contribution to OOS forecasting, providing a new lens through which to evaluate the empirical content of competing asset-pricing models.

Our work speaks to several strands of the literature. The ML asset-pricing literature, initiated by Gu et al. (2020), demonstrates that flexible nonparametric methods substantially outperform linear models in predicting the cross-section of returns. Subsequent work explores neural-network architectures (Chen et al., 2024), autoencoder factor models with no-arbitrage constraints (Gu et al., 2021), conditional factor models (Kelly et al., 2019), and Gaussian process regression applied to equity returns (Filipović and Pasricha, 2022). Kelly et al. (2024) provide a theoretical foundation for this success, showing that complex, overparameterized models are not penalized by benign overfitting and systematically outperform simpler ones. These methods are powerful but agnostic about economic structure. Our contribution is to show that injecting economic theory into the regularization of a kernel method yields both predictive gains and economic interpretability.

A growing body of work recognizes that theory and statistics are complements, not substitutes, for return prediction. The paper closest to ours in spirit is Chen et al. (2026), who show that pre-training neural networks on synthetic data generated by structural models—and then fine-tuning on empirical data—substantially improves option-pricing accuracy, even when the structural model is misspecified. Our approach shares the same fundamental insight—misspecified theory still helps—but differs in three important respects. First, we use soft penalization in the objective function rather than pre-training: structural content enters directly as penalty terms with cross-validated weights, not as an initialization from synthetic data. Second, we target the cross-section of equity returns rather than option pricing. Third, the convexity of our KRR objective guarantees a global optimum and yields tractable asymptotics, whereas neural-network pre-training is inherently non-convex. Most

importantly, our penalty weights provide a continuous measure of each theory's marginal contribution to prediction, whereas transfer learning offers a binary use-or-not decision for each structural model.

A second closely related paper is Bryzgalova et al. (2023a), who develop a Bayesian framework that averages over 2.25 quadrillion linear asset-pricing models formed by subsets of 51 observable factors. Their identification-strength-based priors impose heterogeneous shrinkage—factors that are weakly identified receive stronger shrinkage—and the resulting Bayesian model-averaging stochastic discount factor dominates individual factor models out of sample. Like us, they produce an empirical ranking of economic models. The key differences are the objects being ranked and the estimation framework. Bryzgalova, Huang, and Julliard rank observable factor portfolios within a linear SDF; we rank structural restrictions from named theoretical models within a nonparametric RKHS. Their priors are calibrated from factor identification strength; our penalty weights are selected by temporal cross-validation, tying the ranking directly to out-of-sample predictive performance.

More broadly, the question of which factors or models survive in a high-dimensional setting has been explored from several angles. Kozak et al. (2020) propose economically motivated shrinkage of SDF coefficients proportional to the eigenvalues of characteristic-based principal components, motivated by no-arbitrage bounds on Sharpe ratios. Feng et al. (2020) develop a double-selection LASSO procedure to test whether newly proposed factors have marginal explanatory power beyond existing ones. Harvey et al. (2016) document severe data snooping in factor discovery and argue for higher significance thresholds. Barillas and Shanken (2018) and Kan et al. (2013) provide formal Bayesian and frequentist frameworks for pairwise model comparison. Our paper complements this literature by moving from factor-level selection in linear models to theory-level selection in a nonparametric estimator, where the "theories" are entire families of structural restrictions rather than individual tradable factors.

The econometrics literature on constrained nonparametric estimation provides our math-

ematical foundation. Babii et al. (2024) develop the theory of penalized kernel regression with functional constraints, establishing convergence rates and conditions under which constraints improve estimation. Gabaix et al. (2024) show how demand-system equilibrium conditions can be imposed on asset-pricing estimators, and Bryzgalova et al. (2023b) impose shape restrictions (monotonicity, convexity) on pricing-kernel estimators. In the penalized GMM literature, Caner (2009) and Caner and Zhang (2014) develop LASSO-type and elastic-net GMM estimators with oracle properties for moment selection. Freyberger et al. (2020) use adaptive group LASSO to select characteristics nonparametrically. Our work extends this line by moving from reduced-form shape restrictions to model-implied structural restrictions, and by allowing the data to select among competing restrictions via cross-validation rather than imposing all restrictions with equal weight. The representer theorem, generalized to our multi-penalty setting following Schölkopf and Smola (2002) and the foundational results of Kimeldorf and Wahba (1971) and Wahba (1990), ensures that the solution lies in the span of kernel evaluations even with the added structural penalties.

The paper proceeds as follows. Section 2 presents the mathematical framework, defining the Theory-KRR objective and establishing its properties. Section 3 catalogs the fifty-six structural restrictions we draw from the asset-pricing literature, organized by model class. Section 4 addresses practical estimation, including cross-validation of the multiplier vector $\boldsymbol{\mu}$, computational strategies, and implementation details. Section 5 describes the data and baseline models. Section 6 presents the main empirical results: OOS predictive performance, the theory ranking, and robustness exercises. Section 7 concludes.

2 The Theory-Informed Kernel Ridge Regression Estimator

This section develops the econometric framework. Section 2.1 introduces notation and the prediction environment. Section 2.2 defines the theory-informed KRR objective. Sec-

tions 2.3–2.5 formalize the structural penalty types and establish the representer theorem and finite-dimensional reduction that make the estimator computationally tractable. Section 2.6 shows that the framework nests several influential estimators as special cases. Sections 2.7–2.9 address implementation, misspecification robustness, and inference.

2.1 Setup and Notation

We observe a panel of N assets over T periods. Let $R_{i,t+1}$ denote the excess return of asset i realized at date $t+1$, and let $\mathbf{X}_{i,t} \in \mathbb{R}^p$ collect firm-level characteristics and macroeconomic state variables observable at date t. The total sample size is $n = NT$.

Assumption 1 (Data-generating process)**.** There exists a measurable function $f^* : \mathbb{R}^p \to \mathbb{R}$ such that

$$R_{i,t+1} = f^*(\mathbf{X}_{i,t}) + \varepsilon_{i,t+1}, \qquad \mathbb{E}[\varepsilon_{i,t+1} \mid \mathbf{X}_{i,t}] = 0, \tag{1}$$

where $f^*(\mathbf{X}_{i,t}) = \mathbb{E}[R_{i,t+1} \mid \mathbf{X}_{i,t}]$ is the conditional expected excess return.

We approximate f^* within a reproducing kernel Hilbert space (RKHS).

Definition 2 (RKHS)**.** Let $k : \mathbb{R}^p \times \mathbb{R}^p \to \mathbb{R}$ be a symmetric, positive-definite kernel. The RKHS $\mathcal{H}$ associated with k is the completion of $\operatorname{span}\{k(\mathbf{x}, \cdot) : \mathbf{x} \in \mathbb{R}^p\}$ under the inner product satisfying the reproducing property $f(\mathbf{x}) = \langle f, k(\mathbf{x}, \cdot)\rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$. The RKHS norm $\|f\|_{\mathcal{H}}^2 = \langle f, f\rangle_{\mathcal{H}}$ measures the complexity of f.

We stack observations using the index $s = (i, t)$, so that $\mathbf{X}_s \equiv \mathbf{X}_{i,t}$ and $R_s \equiv R_{i,t+1}$ for $s = 1, \ldots, n$. The $n \times n$ kernel (Gram) matrix is $\mathbf{K}$, with $[\mathbf{K}]_{s,s'} = k(\mathbf{X}_s, \mathbf{X}_{s'})$. The return vector is $\mathbf{R} = (R_1, \ldots, R_n)^\top \in \mathbb{R}^n$.

2.2 The Estimator

Our estimator augments the standard KRR objective with a collection of structural penalty terms, each encoding restrictions implied by a specific economic model.

Definition 3 (Theory-Informed KRR)**.** Let $\{C_j\}_{j=1}^{J}$ be a collection of structural penalty functionals $C_j : \mathcal{H} \to \mathbb{R}_+$, where each $C_j(f) \geq 0$ with equality when f satisfies the restrictions of economic model j. Let $\lambda_{\text{stat}} > 0$ and $\mu_j \geq 0$, $j = 1, \ldots, J$. The *theory-informed KRR* estimator is

$$\hat{f} \;=\; \arg\min_{f \in \mathcal{H}} \underbrace{\frac{1}{n}\sum_{s=1}^{n}\big(R_s - f(\mathbf{X}_s)\big)^2}_{\text{prediction loss}} + \underbrace{\lambda_{\text{stat}}\,\|f\|_{\mathcal{H}}^2}_{\text{statistical regularization}} + \underbrace{\sum_{j=1}^{J}\mu_j\, C_j(f)}_{\text{structural penalties}} . \tag{2}$$

The first two terms constitute standard kernel ridge regression: the squared-error prediction loss trades off against the RKHS-norm penalty, which controls the smoothness of f. The third term is new. Each C_j penalizes deviations from a specific economic model's predictions, and the associated multiplier μ_j governs how strongly that model's restrictions influence the estimate. When $\mu_j = 0$, model j is ignored; as $\mu_j \to \infty$, the restrictions of model j are imposed exactly.

Remark 4 (Interpretation of multipliers). The vector $(\lambda_{\text{stat}}, \mu_1, \ldots, \mu_J)$ balances agnostic statistical regularization against theory-informed regularization. A large λ_{stat} relative to μ_j yields an estimate close to standard KRR; large μ_j relative to λ_{stat} forces the estimator to respect model j's restrictions even if doing so worsens in-sample fit. Cross-validation selects the balance that maximizes out-of-sample predictive accuracy (Section 2.7).

2.3 Structural Penalty Types

We formalize four classes of structural penalties. Each encodes restrictions from a distinct family of asset-pricing models. A key requirement—verified explicitly below—is that every $C_j(f)$ depends on f only through its evaluations at the observed data points $\{f(\mathbf{X}_s)\}_{s=1}^{n}$.

2.3.1 Type A: Euler Equation Penalties

Let model j specify a stochastic discount factor (SDF) $M_{j,t+1} = M_j(\theta_j, \mathbf{X}_t, \mathbf{X}_{t+1})$ parameterized by θ_j, with a preliminary estimator $\hat{\theta}_j$ obtained by GMM or maximum likelihood. The Euler equation requires $\mathbb{E}_t[M_{j,t+1}\, R^{\text{gross}}_{i,t+1}] = 1$ for every asset i and date t, where $R^{\text{gross}}_{i,t+1} = f(\mathbf{X}_{i,t}) + R^f_t$ is the gross return and R^f_t is the risk-free rate.

Definition 5 (Euler equation penalty)**.** For SDF model j with pre-estimated parameter $\hat{\theta}_j$, define

$$C_j^{\text{Euler}}(f) \;=\; \sum_{t=1}^{T} \left\| \frac{1}{N} \sum_{i=1}^{N} M_j\big(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1}\big) \cdot \big(f(\mathbf{X}_{i,t}) + R^f_t\big) - 1 \right\|^2 . \tag{3}$$

The penalty is zero precisely when the cross-sectional Euler equation holds at every date t given $\hat{\theta}_j$. Since $M_j(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1})$ and R^f_t are data, the penalty depends on f only through $\{f(\mathbf{X}_{i,t})\}_{i,t}$.

Remark 6 (Coverage). Type A penalties accommodate consumption-based models (CCAPM, long-run risk, rare disasters, habit formation), as well as production-based SDFs derived from firms' intertemporal optimality conditions. Each distinct SDF specification generates a separate penalty C_j^{Euler}.

2.3.2 Type B: Production and Investment FOC Penalties

Production-based asset pricing relates expected returns to observable investment and profitability variables through firms' first-order conditions. Let model j imply a mapping $g_j : \mathbb{R}^{d_j} \to \mathbb{R}$ from investment characteristics $\mathbf{Z}^{(j)}_{i,t}$ (e.g., investment-to-capital ratio $I_{i,t}/K_{i,t}$, return on equity $\text{ROE}_{i,t}$, asset growth) to expected returns.

Definition 7 (Production FOC penalty)**.** For production model j with predicted returns $g_j(\mathbf{Z}^{(j)}_{i,t})$, define

$$C_j^{\text{prod}}(f) \;=\; \frac{1}{n} \sum_{s=1}^{n} \big(f(\mathbf{X}_s) - g_j\big(\mathbf{Z}^{(j)}_s\big)\big)^2 . \tag{4}$$

The penalty is a mean-squared deviation between the nonparametric estimate and the production model's implied expected return. Since $g_j(\mathbf{Z}_s^{(j)})$ is computed from data, the penalty depends on f only through $\{f(\mathbf{X}_s)\}_{s=1}^n$.

Remark 8 (Nesting the q-theory framework). When g_j encodes the first-order condition from the investment CAPM or the q-factor model, Type B penalties embed the production approach to asset pricing directly into the nonparametric objective.

2.3.3 Type C: Factor Structure Penalties

Factor models impose that expected returns obey a linear pricing relation with respect to exposure to priced factors. Let model j specify a factor $F_{j,t}$ (e.g., an intermediary capital ratio factor, market excess return, or a latent factor). Let $\beta_{i,j}$ denote asset i's loading on $F_{j,t}$, pre-estimated by time-series regression, and let $\alpha_{i,j}$ be the corresponding intercept.

Definition 9 (Factor structure penalty)**.** For factor model j with estimated loadings $\hat{\beta}_{i,j}$, $\hat{\alpha}_{i,j}$ $(i = 1, \ldots, N)$ and factor realizations $F_{j,t}$ $(t = 1, \ldots, T)$, define

$$C_j^{\text{factor}}(f) \;=\; \frac{1}{n}\sum_{i=1}^{N}\sum_{t=1}^{T}\big(f(\mathbf{X}_{i,t}) - \hat{\beta}_{i,j}\,F_{j,t} - \hat{\alpha}_{i,j}\big)^2. \tag{5}$$

When asset pricing restricts $\hat{\alpha}_{i,j} = 0$ for all i (exact factor pricing), the penalty reduces to $\frac{1}{n}\sum_{i,t}(f(\mathbf{X}_{i,t}) - \hat{\beta}_{i,j}\,F_{j,t})^2$, penalizing deviations from the factor model's predictions of expected returns. Again, the penalty depends on f only through $\{f(\mathbf{X}_{i,t})\}_{i,t}$.

Remark 10 (Intermediary asset pricing). When $F_{j,t}$ is the intermediary capital ratio or the intermediary leverage factor, Type C penalties embed intermediary-based models into the nonparametric estimator. This is particularly valuable when the intermediary model captures time-variation in risk premia that firm-level characteristics alone cannot.

2.3.4 Type D: Demand-System Equilibrium Penalties

Demand-system approaches to asset pricing derive equilibrium prices from investors' portfolio demand functions. Let $D_i : \mathbb{R}^N \to \mathbb{R}^N$ map the vector of expected returns to investor i's optimal portfolio weights, and let $w_i > 0$ denote investor i's wealth share, with the index now running over investors rather than assets. Let $\mathbf{S}_t \in \mathbb{R}^N$ denote the vector of asset market-capitalization shares (the supply side) at date t.

Definition 11 (Demand-system equilibrium penalty)**.** For demand-system model j with demand functions $\{D_i^{(j)}\}$ and observed supplies $\{\mathbf{S}_t\}$, let $\hat{\mathbf{f}}_t = (f(\mathbf{X}_{1,t}), \ldots, f(\mathbf{X}_{N,t}))^\top$ collect expected-return evaluations. Define

$$C_j^{\text{demand}}(f) \;=\; \sum_{t=1}^{T} \left\| \sum_i w_i^{(j)} \, D_i^{(j)}\big(\hat{\mathbf{f}}_t\big) - \mathbf{S}_t \right\|^2 . \tag{6}$$

The penalty measures the squared market-clearing failure: it is zero when the demand system clears at the predicted expected returns. It depends on f only through $\{f(\mathbf{X}_{i,t})\}_{i,t}$.

Remark 12 (Connections to the demand-system literature)*.* The demand-system approach models institutional investors' demand as a function of characteristics and expected returns. Type D penalties encode the equilibrium restriction that aggregate demand equals supply, translating the demand-system estimator's logic into a penalty within our nonparametric framework. When $D_i^{(j)}$ is linear in expected returns, C_j^{demand} is quadratic in $\hat{\mathbf{f}}_t$ and the overall problem remains convex.

2.4 The Representer Theorem

The following result establishes that the infinite-dimensional optimization problem (2) admits a finite-dimensional solution, despite the presence of the structural penalties.

Proposition 13 (Representer theorem for theory-informed KRR)**.** *Let $\mathcal{H}$ be an RKHS with kernel k. Suppose each structural penalty $C_j : \mathcal{H} \to \mathbb{R}_+$ depends on f only through the vector*

of evaluations $(f(\mathbf{X}_1), \ldots, f(\mathbf{X}_n))^\top \in \mathbb{R}^n$. *That is, there exist functions* $\tilde{C}_j : \mathbb{R}^n \to \mathbb{R}_+$ *such that*

$$C_j(f) = \tilde{C}_j\big(f(\mathbf{X}_1), \ldots, f(\mathbf{X}_n)\big) \qquad \text{for all } f \in \mathcal{H},\ j = 1, \ldots, J. \tag{7}$$

Then any minimizer of (2) *admits the representation*

$$\hat{f}(\cdot) = \sum_{s=1}^{n} \hat{\alpha}_s\, k(\mathbf{X}_s, \cdot), \tag{8}$$

for some $\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \ldots, \hat{\alpha}_n)^\top \in \mathbb{R}^n$.

Proof. Decompose any $f \in \mathcal{H}$ as $f = f_\parallel + f_\perp$, where $f_\parallel$ lies in the finite-dimensional subspace $\mathcal{V} = \operatorname{span}\{k(\mathbf{X}_s, \cdot) : s = 1, \ldots, n\}$ and $f_\perp \in \mathcal{V}^\perp$. By the reproducing property,

$$f(\mathbf{X}_s) = \langle f, k(\mathbf{X}_s, \cdot)\rangle_{\mathcal{H}} = \langle f_\parallel, k(\mathbf{X}_s, \cdot)\rangle_{\mathcal{H}} + \langle f_\perp, k(\mathbf{X}_s, \cdot)\rangle_{\mathcal{H}} = f_\parallel(\mathbf{X}_s),$$

since $k(\mathbf{X}_s, \cdot) \in \mathcal{V}$ and $f_\perp \perp \mathcal{V}$. Therefore the prediction loss $\frac{1}{n}\sum_s (R_s - f(\mathbf{X}_s))^2$ depends only on $f_\parallel$, and so does every $C_j(f)$ by assumption (7).

For the RKHS norm, orthogonality gives $\|f\|_{\mathcal{H}}^2 = \|f_\parallel\|_{\mathcal{H}}^2 + \|f_\perp\|_{\mathcal{H}}^2$. Since $\|f_\perp\|_{\mathcal{H}}^2 \geq 0$ and $\lambda_{\text{stat}} > 0$, the objective is strictly decreased by setting $f_\perp = 0$ whenever $f_\perp \neq 0$. Hence any minimizer satisfies $f = f_\parallel \in \mathcal{V}$, which yields the representation (8). □

Remark 14 (Verification for Types A–D). The condition (7) holds for all four penalty types:

(a) *Type A (Euler):* $C_j^{\text{Euler}}(f)$ depends on f through $\{f(\mathbf{X}_{i,t})\}_{i,t}$, since $M_j(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1})$ and R_t^f are data.

(b) *Type B (Production):* $C_j^{\text{prod}}(f)$ is explicitly a function of $\{f(\mathbf{X}_s)\}_s$ and data $\{g_j(\mathbf{Z}_s^{(j)})\}_s$.

(c) *Type C (Factor):* $C_j^{\text{factor}}(f)$ involves $\{f(\mathbf{X}_{i,t})\}_{i,t}$ and pre-estimated $\hat{\beta}_{i,j}$, $\hat{\alpha}_{i,j}$, $F_{j,t}$.

(d) *Type D (Demand):* $C_j^{\text{demand}}(f)$ depends on f through $\hat{\mathbf{f}}_t = (f(\mathbf{X}_{1,t}), \ldots, f(\mathbf{X}_{N,t}))^\top$.

In each case, the penalty is a function of the n-dimensional evaluation vector, so Proposition 13 applies. This extends classical representer theorems to accommodate theory-informed structural penalties.

2.5 Finite-Dimensional Reduction

By Proposition 13, it suffices to optimize over $\boldsymbol{\alpha} \in \mathbb{R}^n$. Substituting $f(\cdot) = \sum_s \alpha_s\, k(\mathbf{X}_s, \cdot)$ into the objective yields the following.

Proposition 15 (Finite-dimensional problem)**.** *Under the representer theorem (Proposition 13), the theory-informed KRR estimator* (2) *reduces to*

$$\hat{\boldsymbol{\alpha}} \;=\; \arg\min_{\boldsymbol{\alpha}\in\mathbb{R}^n} \; \frac{1}{n}\|\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}\|^2 \;+\; \lambda_{\text{stat}}\, \boldsymbol{\alpha}^\top \mathbf{K}\, \boldsymbol{\alpha} \;+\; \sum_{j=1}^{J} \mu_j\, \tilde{C}_j(\mathbf{K}\boldsymbol{\alpha}), \tag{9}$$

where $\mathbf{K}$ *is the* $n \times n$ *kernel matrix,* $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$ *is the* $n \times 1$ *vector of fitted values, and* $\tilde{C}_j(\hat{\mathbf{f}}) = C_j(f)$ *expresses each penalty in terms of fitted values.*

Proof. The prediction loss becomes $\frac{1}{n}\|\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}\|^2$ since $f(\mathbf{X}_s) = \sum_{s'} \alpha_{s'} k(\mathbf{X}_{s'}, \mathbf{X}_s) = [\mathbf{K}\boldsymbol{\alpha}]_s$. For the RKHS norm,

$$\|f\|_{\mathcal{H}}^2 = \left\langle \sum_s \alpha_s\, k(\mathbf{X}_s, \cdot),\; \sum_{s'} \alpha_{s'}\, k(\mathbf{X}_{s'}, \cdot) \right\rangle_{\mathcal{H}} = \sum_{s,s'} \alpha_s\, \alpha_{s'}\, k(\mathbf{X}_s, \mathbf{X}_{s'}) = \boldsymbol{\alpha}^\top \mathbf{K}\, \boldsymbol{\alpha}.$$

By condition (7), each $C_j(f)$ is a function of $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$. Combining yields (9). □

2.5.1 Convexity and First-Order Conditions for Quadratic Penalties

When all active structural penalties are quadratic in the fitted values $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$, the finite-dimensional problem (9) becomes a convex quadratic program with a closed-form solution.

For Type A penalties, define the $T \times n$ matrix $\mathbf{A}_j$ with rows indexed by t:

$$[\mathbf{A}_j]_{t,s} = \begin{cases} \dfrac{1}{N} M_j\big(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1}\big) & \text{if } s = (i,t) \text{ for some } i, \\ 0 & \text{otherwise,} \end{cases} \tag{10}$$

and the $T \times 1$ vector $\mathbf{b}_j$ with $[\mathbf{b}_j]_t = 1 - \frac{1}{N}\sum_i M_j(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1})\, R_t^f$. Then

$$C_j^{\text{Euler}}(f) = \|\mathbf{A}_j\, \hat{\mathbf{f}} - \mathbf{b}_j\|^2 = (\mathbf{A}_j\, \mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j)^\top (\mathbf{A}_j\, \mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j). \tag{11}$$

For Type B penalties, let $\mathbf{g}_j \in \mathbb{R}^n$ with $[\mathbf{g}_j]_s = g_j(\mathbf{Z}_s^{(j)})$. Then

$$C_j^{\text{prod}}(f) = \frac{1}{n}\|\mathbf{K}\boldsymbol{\alpha} - \mathbf{g}_j\|^2. \tag{12}$$

For Type C penalties, let $\mathbf{h}_j \in \mathbb{R}^n$ with $[\mathbf{h}_j]_s = \hat{\beta}_{i,j}\, F_{j,t} + \hat{\alpha}_{i,j}$ where $s = (i,t)$. Then

$$C_j^{\text{factor}}(f) = \frac{1}{n}\|\mathbf{K}\boldsymbol{\alpha} - \mathbf{h}_j\|^2. \tag{13}$$

Proposition 16 (First-order conditions)**.** *Suppose every active penalty ($\mu_j > 0$) is of Type A, B, or C (or more generally, quadratic in $\hat{\mathbf{f}}$). Define the aggregated penalty matrix*

$$\boldsymbol{\Omega} \;=\; \sum_{j \in \mathcal{J}^A} \mu_j\, \mathbf{A}_j^\top \mathbf{A}_j \;+\; \sum_{j \in \mathcal{J}^B} \frac{\mu_j}{n}\, \mathbf{I}_n \;+\; \sum_{j \in \mathcal{J}^C} \frac{\mu_j}{n}\, \mathbf{I}_n, \tag{14}$$

where $\mathcal{J}^A$, $\mathcal{J}^B$, $\mathcal{J}^C$ index Type A, B, and C penalties respectively, and the aggregated target vector

$$\boldsymbol{\omega} \;=\; \sum_{j \in \mathcal{J}^A} \mu_j\, \mathbf{A}_j^\top \mathbf{b}_j \;+\; \sum_{j \in \mathcal{J}^B} \frac{\mu_j}{n}\, \mathbf{g}_j \;+\; \sum_{j \in \mathcal{J}^C} \frac{\mu_j}{n}\, \mathbf{h}_j. \tag{15}$$

Then the first-order conditions for (9) are

$$\left[\frac{1}{n}\,\mathbf{K} + \lambda_{\text{stat}}\, \mathbf{I}_n + \boldsymbol{\Omega}\,\mathbf{K}\right] \boldsymbol{\alpha} \;=\; \frac{1}{n}\,\mathbf{R} + \boldsymbol{\omega}, \tag{16}$$

and when $\mathbf{K}$ *is strictly positive definite, the unique solution is*

$$\hat{\boldsymbol{\alpha}} \;=\; \left[\frac{1}{n}\,\mathbf{K} + \lambda_{\text{stat}}\,\mathbf{I}_n + \boldsymbol{\Omega}\,\mathbf{K}\right]^{-1}\left(\frac{1}{n}\,\mathbf{R} + \boldsymbol{\omega}\right). \tag{17}$$

Proof. Write the objective (9) with quadratic penalties as

$$\begin{aligned} L(\boldsymbol{\alpha}) = \frac{1}{n}(\mathbf{R} - \mathbf{K}\boldsymbol{\alpha})^\top(\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}) + \lambda_{\text{stat}}\,\boldsymbol{\alpha}^\top\mathbf{K}\boldsymbol{\alpha} \\ + \sum_{j\in\mathcal{J}^A}\mu_j(\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j)^\top(\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j) + \sum_{j\in\mathcal{J}^{B,C}}\frac{\mu_j}{n}(\mathbf{K}\boldsymbol{\alpha} - \mathbf{v}_j)^\top(\mathbf{K}\boldsymbol{\alpha} - \mathbf{v}_j), \end{aligned} \tag{18}$$

where $\mathbf{v}_j = \mathbf{g}_j$ for Type B and $\mathbf{v}_j = \mathbf{h}_j$ for Type C. Taking the gradient with respect to $\boldsymbol{\alpha}$:

$$\begin{aligned} \frac{\partial L}{\partial \boldsymbol{\alpha}} = -\frac{2}{n}\,\mathbf{K}(\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}) + 2\lambda_{\text{stat}}\,\mathbf{K}\boldsymbol{\alpha} + 2\sum_{j\in\mathcal{J}^A}\mu_j\,\mathbf{K}\mathbf{A}_j^\top(\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j) \\ + 2\sum_{j\in\mathcal{J}^{B,C}}\frac{\mu_j}{n}\,\mathbf{K}(\mathbf{K}\boldsymbol{\alpha} - \mathbf{v}_j). \end{aligned} \tag{19}$$

Setting (19) to zero and left-multiplying by $\mathbf{K}^{-1}$ (which exists when $\mathbf{K} \succ 0$):

$$\frac{1}{n}\,\mathbf{K}\boldsymbol{\alpha} + \lambda_{\text{stat}}\,\boldsymbol{\alpha} + \sum_{j\in\mathcal{J}^A}\mu_j\,\mathbf{A}_j^\top\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} + \sum_{j\in\mathcal{J}^{B,C}}\frac{\mu_j}{n}\,\mathbf{K}\boldsymbol{\alpha} = \frac{1}{n}\,\mathbf{R} + \sum_{j\in\mathcal{J}^A}\mu_j\,\mathbf{A}_j^\top\mathbf{b}_j + \sum_{j\in\mathcal{J}^{B,C}}\frac{\mu_j}{n}\,\mathbf{v}_j.$$

Collecting terms using the definitions of $\boldsymbol{\Omega}$ and $\boldsymbol{\omega}$ from (14)–(15) gives (16). Strict positive definiteness of $\mathbf{K}$ and $\lambda_{\text{stat}} > 0$ ensure the coefficient matrix in (16) is invertible, yielding uniqueness. □

Remark 17 (Computational cost). The dominant cost is solving the $n \times n$ linear system (17), which requires $O(n^3)$ operations. When n is large, one can exploit the eigendecomposition $\mathbf{K} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ (computed once at cost $O(n^3)$) and solve in the rotated basis, where the matrix becomes diagonal plus a low-rank perturbation from the Type A penalties. Alternatively, random Fourier features or Nyström approximations reduce the effective dimension.

Remark 18 (Type D penalties and convexity). When demand functions $D_i^{(j)}$ are linear in

expected returns, C_j^{demand} is quadratic in $\hat{\mathbf{f}}$ and the problem remains a convex QP with a structure analogous to (16). For nonlinear demand systems, the problem is generally nonconvex and requires iterative methods (e.g., projected gradient descent or alternating minimization), though convexity of the remaining terms provides a useful warm start.

2.6 Nesting Structure

A key advantage of the framework is that it nests several influential estimators as special or limiting cases. This both clarifies the relationship between existing methods and suggests that the theory-informed KRR estimator can weakly dominate each special case.

Proposition 19 (Nesting)**.** *The theory-informed KRR estimator* (2) *nests the following:*

(i) ***Standard KRR.*** *Setting $\mu_j = 0$ for all $j = 1, \ldots, J$ yields*

$$\hat{f}^{\text{KRR}} = \arg\min_{f \in \mathcal{H}} \frac{1}{n} \sum_s (R_s - f(\mathbf{X}_s))^2 + \lambda_{\text{stat}} \|f\|_{\mathcal{H}}^2,$$

the standard kernel ridge regression estimator.

(ii) ***Constrained minimum-distance / GMM.*** *Setting $\lambda_{\text{stat}} = 0$ and activating a single Type A penalty with $\mu_j \to \infty$ yields*

$$\hat{f}^{\text{GMM}} = \arg\min_{f \in \mathcal{H}} \frac{1}{n} \sum_s (R_s - f(\mathbf{X}_s))^2 \quad s.t. \quad C_j^{\text{Euler}}(f) = 0,$$

which is a nonparametric generalization of GMM-based SDF estimation. When $\mathcal{H}$ is restricted to parametric functions, this recovers classical GMM.

(iii) ***Approximate instrumented PCA.*** *Consider a linear kernel $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$ so that $f(\mathbf{X}_{i,t}) = \boldsymbol{\beta}^\top \mathbf{X}_{i,t}$ for some $\boldsymbol{\beta} \in \mathbb{R}^p$, and a Type C penalty where the factor $F_{j,t}$ is*

a target-PCA factor. Then (2) becomes

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{n} \sum_{i,t} (R_{i,t+1} - \boldsymbol{\beta}^\top \mathbf{X}_{i,t})^2 + \lambda_{\text{stat}} \|\boldsymbol{\beta}\|^2 + \mu_j \frac{1}{n} \sum_{i,t} (\boldsymbol{\beta}^\top \mathbf{X}_{i,t} - \hat{\beta}_{i,j} F_{j,t})^2, \quad (20)$$

a ridge regression with an additional penalty encouraging alignment with the target-PCA factor structure. This approximates the cross-sectional and time-series restrictions in instrumented PCA methods.

(iv) ***Text-informed penalized regression.*** *Replacing $\mathcal{H}$ with a finite-dimensional linear space $\mathbb{R}^p$, substituting the RKHS-norm penalty with an ℓ_1 norm $\|\boldsymbol{\beta}\|_1$, and setting $C_j(\boldsymbol{\beta}) = \sum_{k=1}^p \gamma_{j,k} |\beta_k|$ where $\gamma_{j,k}$ are text-derived importance weights, yields*

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{n} \sum_{s} (R_s - \boldsymbol{\beta}^\top \mathbf{X}_s)^2 + \lambda_{\text{stat}} \|\boldsymbol{\beta}\|_1 + \mu_j \sum_{k} \gamma_{j,k} |\beta_k|, \quad (21)$$

a variant of the adaptive Lasso with text-informed variable-specific penalties, nesting the LLM-Lasso approach of incorporating domain knowledge from large language models into penalized regression.

Proof. Claims (i) and (ii) follow directly from the definition. For (i), setting $\mu_j = 0$ eliminates the structural penalties. For (ii), taking $\mu_j \to \infty$ enforces $C_j(f) = 0$ as a constraint; formally, the sequence of penalized minimizers converges to the constrained minimizer provided C_j is lower semicontinuous and $\{f \in \mathcal{H} : C_j(f) = 0\}$ is nonempty. Setting $\lambda_{\text{stat}} = 0$ removes agnostic regularization.

For (iii), the linear kernel restricts $\mathcal{H}$ to linear functions $f(\mathbf{x}) = \boldsymbol{\beta}^\top \mathbf{x}$, with $\|f\|_{\mathcal{H}}^2 = \|\boldsymbol{\beta}\|^2$. Substituting a Type C penalty with target-PCA factor loadings and simplifying gives (20).

For (iv), the correspondence is notational: the ℓ_1 norm replaces the ℓ_2 RKHS norm, and the structural penalty becomes a weighted ℓ_1 term on coefficients rather than a penalty on function evaluations. The estimator (21) coincides with the adaptive Lasso when $\gamma_{j,k} = |\hat{\beta}_k^{\text{init}}|^{-1}$ and with text-informed penalization when $\gamma_{j,k}$ reflects LLM-derived variable rele-

vance scores. □

Remark 20 (Weak dominance). Since standard KRR corresponds to $\mu_j = 0$ for all j, and the cross-validated theory-informed KRR is free to select $\hat{\mu}_j = 0$ if theory j is unhelpful, the theory-informed estimator weakly dominates standard KRR in out-of-sample prediction loss—up to the granularity of the hyperparameter search. The same logic applies to each nested special case: the larger hyperparameter space can only improve (or match) the best achievable out-of-sample performance.

2.7 Cross-Validation for Hyperparameter Selection

The hyperparameter vector $\boldsymbol{\lambda} = (\lambda_{\text{stat}}, \mu_1, \ldots, \mu_J)$ is $(J+1)$-dimensional. We introduce grouped penalties and temporal cross-validation to make the selection tractable.

Grouped penalties. We assign each model j to one of G groups based on its economic type. Let $g : \{1, \ldots, J\} \to \{1, \ldots, G\}$ denote the group assignment, and impose $\mu_j = \mu_{g(j)}$ for all j. In our empirical application, $G = 8$, corresponding to the four penalty types (A–D) each potentially with two variants (e.g., consumption-based vs. production-based SDFs within Type A). This reduces the hyperparameter space from $(J+1)$ to $(G+1) = 9$ dimensions.

Temporal cross-validation. Financial panel data exhibit strong serial dependence, so we use leave-one-year-out (LOYO) cross-validation to select $\boldsymbol{\lambda}$. Partition the sample into T_{yr} calendar years $\mathcal{T}_1, \ldots, \mathcal{T}_{T_{\text{yr}}}$. For fold ℓ, let $\mathcal{S}_{-\ell} = \{(i,t) : t \notin \mathcal{T}_\ell\}$ be the training set and $\mathcal{S}_\ell = \{(i,t) : t \in \mathcal{T}_\ell\}$ the validation set. The cross-validated objective is

$$\text{CV}(\boldsymbol{\lambda}) \;=\; \frac{1}{T_{\text{yr}}} \sum_{\ell=1}^{T_{\text{yr}}} \frac{1}{|\mathcal{S}_\ell|} \sum_{s \in \mathcal{S}_\ell} \big(R_s - \hat{f}_{-\ell}(\mathbf{X}_s; \boldsymbol{\lambda})\big)^2, \tag{22}$$

where $\hat{f}_{-\ell}(\cdot;\boldsymbol{\lambda})$ is the theory-informed KRR estimator trained on $\mathcal{S}_{-\ell}$ with hyperparameters $\boldsymbol{\lambda}$.

Search strategy. We minimize $\mathrm{CV}(\boldsymbol{\lambda})$ by random search over a 9-dimensional log-uniform grid: $\log_{10}\lambda_{\text{stat}} \in [-6,2]$ and $\log_{10}\mu_g \in [-6,2]$ for $g = 1,\ldots,G$. We draw $B = 500$ random configurations and evaluate each by solving the finite-dimensional problem (9) T_{yr} times (once per fold). The selected hyperparameters are $\hat{\boldsymbol{\lambda}} = \arg\min_{\boldsymbol{\lambda}} \mathrm{CV}(\boldsymbol{\lambda})$.

Remark 21 (Computational considerations). Each fold requires solving a linear system of size $|\mathcal{S}_{-\ell}|$, which is $O(((T_{\text{yr}}-1)\cdot N/T_{\text{yr}})^3)$ per fold. With $T_{\text{yr}} \approx 30$–60 years and $N \approx 3{,}000$–10,000 assets, this is demanding but feasible with precomputed kernel eigendecompositions and parallelization across folds and random draws. The total cost scales as $O(B \cdot T_{\text{yr}} \cdot n_{\text{fold}}^3)$ where $n_{\text{fold}} = |\mathcal{S}_{-\ell}|$.

2.8 Misspecification Robustness

A central concern with theory-informed estimation is that imposing restrictions from a misspecified model may worsen predictions. The following result shows that cross-validation provides an automatic safeguard.

Proposition 22 (Misspecification robustness)**.** *Suppose model j is misspecified in the sense that the true conditional expected return function f^* satisfies $C_j(f^*) > 0$. Under the following regularity conditions:*

(R1) *The RKHS $\mathcal{H}$ is rich enough that $\inf_{f\in\mathcal{H}} \|f - f^*\|_{L^2} = 0$;*

(R2) *The cross-validation procedure (22) is asymptotically oracle-efficient in the sense that $\mathrm{CV}(\hat{\boldsymbol{\lambda}}) - \inf_{\boldsymbol{\lambda}} \mathrm{CV}(\boldsymbol{\lambda}) = o_p(1)$;*

(R3) *The penalty C_j is continuous in $\|\cdot\|_{L^2}$ and the parameter space for μ_j includes zero;*

the cross-validated multiplier satisfies $\hat{\mu}_j \xrightarrow{p} 0$ as $n \to \infty$.

Proof. We show that any $\mu_j > 0$ introduces an asymptotic bias, so the oracle sets $\mu_j^* = 0$.

Step 1: Asymptotic bias from misspecification. When $C_j(f^*) > 0$, the set $\mathcal{F}_j(\mu_j) = \arg\min_{f\in\mathcal{H}} \frac{1}{n}\sum_s (R_s - f(\mathbf{X}_s))^2 + \lambda_{\text{stat}}\|f\|_{\mathcal{H}}^2 + \mu_j C_j(f)$ converges (as $n \to \infty$ and $\lambda_{\text{stat}} \to 0$ at an appropriate rate) to the population minimizer

$$f_j^*(\mu_j) = \arg\min_f \mathbb{E}[(R - f(\mathbf{X}))^2] + \mu_j C_j(f).$$

By (R1), $f_j^*(0) = f^*$. For $\mu_j > 0$, the term $\mu_j C_j(f)$ tilts the solution toward $\{f : C_j(f) \text{ is small}\}$. Since $C_j(f^*) > 0$, this tilt is binding: $f_j^*(\mu_j) \neq f^*$ for all $\mu_j > 0$, and the prediction risk satisfies

$$\mathbb{E}\big[(R - f_j^*(\mu_j)(\mathbf{X}))^2\big] > \mathbb{E}\big[(R - f^*(\mathbf{X}))^2\big] = \mathbb{E}[\varepsilon^2] \qquad \text{for all } \mu_j > 0.$$

Step 2: Risk gap is bounded away from zero. By continuity of C_j (condition (R3)) and strict convexity of the population prediction risk, for any $\delta > 0$ there exists $c(\delta) > 0$ such that

$$\text{Risk}(\mu_j) - \text{Risk}(0) \;\geq\; c(\delta) > 0 \qquad \text{for all } \mu_j \geq \delta,$$

where $\text{Risk}(\mu_j) = \mathbb{E}[(R - f_j^*(\mu_j)(\mathbf{X}))^2]$.

Step 3: Oracle efficiency implies $\hat{\mu}_j \to 0$. By (R2), $\text{CV}(\hat{\boldsymbol{\lambda}})$ tracks the oracle $\inf_{\boldsymbol{\lambda}} \text{CV}(\boldsymbol{\lambda})$ up to $o_p(1)$. Since the oracle selects $\mu_j^* = 0$ (the unique risk minimizer along the μ_j coordinate by Step 2), and the risk gap is bounded below by $c(\delta)$ for $\mu_j \geq \delta$, we have $\Pr(\hat{\mu}_j \geq \delta) \to 0$ for every $\delta > 0$. □

Remark 23 (Practical implication). Proposition 22 provides a theoretical guarantee that misspecified theories "turn themselves off" through cross-validation. In finite samples, a moderately misspecified theory may still receive positive $\hat{\mu}_j$ if it captures useful structure despite not being exactly correct. The estimator thus exploits partial truths—a theory need not be exactly right to be useful for prediction.

Remark 24 (Comparison to Bayesian model averaging). The misspecification robustness of theory-informed KRR is analogous to Bayesian model averaging (BMA) in spirit: both frameworks downweight models that fit poorly out of sample. However, the mechanism differs. BMA assigns posterior probabilities to discrete model specifications, whereas theory-informed KRR treats each model's restrictions as a continuous penalty with cross-validated strength. The latter avoids the combinatorial explosion of the BMA model space and allows partial incorporation of each theory's restrictions.

2.9 Inference on Model Value

Beyond prediction, one may wish to test whether a particular economic model contributes statistically significant information beyond agnostic regularization. This amounts to testing

$$H_0 : \mu_j^* = 0 \qquad \text{vs.} \qquad H_1 : \mu_j^* > 0, \tag{23}$$

where μ_j^* is the population-optimal multiplier for model j. Rejection of H_0 implies that model j provides marginal predictive value beyond what the kernel and other models already capture.

2.9.1 Connection to Uniform Inference for KRR

Recent work by Singh and Vijaykumar (2023) establishes uniform confidence bands for kernel ridge regression estimators of conditional expectations. Their framework provides pointwise and uniform inference for $f^*(\mathbf{x})$ across $\mathbf{x} \in \mathcal{X}$ using a bias-corrected estimator and variance estimation based on the kernel's spectral properties.

We outline how this can be extended to test (23).

Proposition 25 (Testing the marginal value of economic models)**.** *Suppose the conditions of Singh and Vijaykumar (2023) hold for the RKHS $\mathcal{H}$, augmented with the following:*

(T1) *The mapping* $\mu_j \mapsto \hat{f}(\cdot\,;\mu_j)$ *is differentiable in* L^2*, with derivative* $\partial_{\mu_j}\hat{f}$ *satisfying* $\|\partial_{\mu_j}\hat{f}\|_{L^2} = O(1)$*;*

(T2) *The estimation error satisfies* $\|\hat{f}(\cdot\,;\hat{\boldsymbol{\lambda}}) - f^*\|_\infty = O_p(r_n)$ *for a rate* $r_n \to 0$*.*

Define the test statistic

$$T_j \;=\; n \cdot \big[\mathrm{CV}(\hat{\boldsymbol{\lambda}}_{-j}) - \mathrm{CV}(\hat{\boldsymbol{\lambda}})\big], \tag{24}$$

where $\hat{\boldsymbol{\lambda}}_{-j}$ *denotes the cross-validated hyperparameters with* μ_j *constrained to zero. Under* $H_0\colon \mu_j^* = 0$*,* $T_j = o_p(1)$*. Under* $H_1\colon \mu_j^* > 0$*,* $T_j \to \infty$ *in probability.*

Proof. Under H_0*.* When $\mu_j^* = 0$, the oracle solution does not use model j. Constraining $\mu_j = 0$ therefore imposes no binding restriction on the population objective. The difference $\mathrm{CV}(\hat{\boldsymbol{\lambda}}_{-j}) - \mathrm{CV}(\hat{\boldsymbol{\lambda}})$ arises solely from finite-sample noise in the cross-validated selection of the remaining hyperparameters. By condition (T2) and the oracle efficiency in (R2), this difference is $O_p(n^{-1}r_n)$, so $T_j = n \cdot O_p(n^{-1}r_n) = O_p(r_n) = o_p(1)$.

Under H_1*.* When $\mu_j^* > 0$, constraining $\mu_j = 0$ removes a binding restriction. By condition (T1), the population risk increases by at least $c > 0$ (a positive constant depending on μ_j^* and C_j). By uniform convergence of the cross-validated risk to the population risk, $\mathrm{CV}(\hat{\boldsymbol{\lambda}}_{-j}) - \mathrm{CV}(\hat{\boldsymbol{\lambda}}) \geq c/2 > 0$ with probability approaching one. Hence $T_j = n(c/2 + o_p(1)) \to \infty$. □

Remark 26 (Permutation inference). A full distributional theory for T_j under H_0 (critical values, size control, power analysis) requires characterizing the joint distribution of the cross-validated hyperparameters, which we defer to future work. In practice, we implement permutation-based inference: randomly reassign the structural penalty C_j to pseudo-models (scrambling the economic content while preserving the penalty's functional form) and compare T_j to the permutation distribution. This yields valid p-values under exchangeability of the pseudo-model assignments.

Remark 27 (Interpretation). The test (23) asks a fundamentally different question from testing whether a model's pricing errors are zero. A model may have large pricing errors

$(C_j(f^*) > 0)$ yet still receive $\mu_j^* > 0$ if its restrictions are correlated with the true f^* in a way that improves finite-sample estimation. Conversely, a correctly specified model $(C_j(f^*) = 0)$ may have $\mu_j^* = 0$ if the RKHS and data are rich enough that the model's restrictions are redundant. The test thus measures the *marginal predictive value* of each theory in the presence of all others.

Remark 28 (Relation to model confidence sets). The pairwise comparison underlying (24) connects to the model confidence set (MCS) framework of Hansen, Lunde, and Nason (2011). One could construct an MCS of theories whose marginal contributions are statistically indistinguishable: $\widehat{\mathrm{MCS}}_\alpha = \{j : T_j \leq c_\alpha\}$, where c_α is calibrated by the permutation distribution. This provides a principled way to report which theories are "useful" for asset-pricing prediction at a given confidence level.

2.10 Summary of the Framework

Table 1 collects the main components of the theory-informed KRR estimator.

Table 1: Summary of the theory-informed KRR framework

Component	**Role**	**Reference**
Prediction loss	Fit to observed returns	Eq. (2)
RKHS penalty	Agnostic statistical regularization	Eq. (2)
Type A penalty	Euler equation restrictions (SDF models)	Def. 5
Type B penalty	Production/investment FOC restrictions	Def. 7
Type C penalty	Factor structure restrictions	Def. 9
Type D penalty	Demand-system equilibrium restrictions	Def. 11
Representer theorem	Finite-dimensional reduction	Prop. 13
First-order conditions	Closed-form for quadratic penalties	Prop. 16
Nesting	Recovers KRR, GMM, IPCA, Lasso variants	Prop. 19
CV selection	Grouped penalties, temporal LOYO	Eq. (22)
Misspecification	Misspecified theories auto-downweight	Prop. 22
Inference	Test marginal value of each theory	Prop. 25

3 Structural Restrictions Catalog

Each restriction in our framework originates from a specific economic model and imposes a distinct functional constraint on the mapping from observable characteristics to expected returns. A consumption-based Euler equation with power utility implies a different penalty $C_j(f)$ than one with Epstein–Zin preferences, even though both restrict the same SDF moment condition, because the functional form of the stochastic discount factor $M_{j,t+1}$ differs. Similarly, a production-based investment CAPM penalty differs from a q-theory penalty not merely in its parameter values but in which investment variables enter the mapping g_j and how they interact. The penalties are not interchangeable: each encodes a qualitatively different economic hypothesis about why expected returns vary across assets and over time.

We organize the fifty-six restrictions into eight families, grouped by economic mechanism: consumption-based Euler equations, production-based and investment restrictions, intermediary and institutional restrictions, information and learning restrictions, demand-system and market microstructure restrictions, variance and volatility restrictions, behavioral and sentiment restrictions, and macro-finance and term structure restrictions. For each restriction, we specify three elements: the source model (the economic theory that implies it), the mathematical form of the restriction (the SDF, first-order condition, or equilibrium condition), and the penalty functional $C_j(f)$ through which it enters the Theory-KRR objective (2) from Section 2. Because every penalty $C_j(f)$ depends on f only through its evaluations at observed data points $\{f(\mathbf{X}_s)\}_{s=1}^n$, the representer theorem (Proposition 13) applies to the full collection, and the estimator remains computationally tractable regardless of how many restrictions are active.

3.1 Consumption-Based Euler Equations

Consumption-based models derive asset-pricing restrictions from the representative agent's intertemporal optimality condition. The Euler equation $\mathbb{E}[M_{t+1}R_{i,t+1}^{\text{gross}} \mid \mathcal{F}_t] = 1$ must hold

for every asset i and date t, where the SDF M_{t+1} is determined by the agent's preferences and the consumption process. Different preference specifications—power utility, recursive utility, habit formation, long-run risk—yield different SDFs and hence different penalties. All thirteen restrictions in this family are Type A (Euler equation) penalties as defined in Section 2.3.1:

$$C_j^{\text{Euler}}(f) \;=\; \sum_{t=1}^{T} \left(\frac{1}{N} \sum_{i=1}^{N} M_j\big(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1}\big) \cdot \big(f(\mathbf{X}_{i,t}) + R_t^f\big) - 1 \right)^2, \tag{25}$$

where $\hat{\theta}_j$ denotes pre-estimated preference parameters obtained by GMM. What distinguishes the thirteen restrictions is solely the form of M_j.

Table 2: Consumption-Based Euler Equation Restrictions

#	Restriction	Source	SDF / Penalty Form
1	Power utility (C-CAPM)	(Breeden, 1979; Lucas, 1978)	$M_{t+1} = \delta \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}$
2	Epstein–Zin recursive utility	(Epstein and Zin, 1989)	$M_{t+1} = \delta^{\theta} \left(\frac{C_{t+1}}{C_t}\right)^{-\theta/\psi} R_{m,t+1}^{\theta-1}, \quad \theta = \frac{1-\gamma}{1-1/\psi}$
3	External habit (surplus ratio)	(Campbell and Cochrane, 1999)	$M_{t+1} = \delta \left(\frac{S_{t+1}}{S_t}\right)^{-\gamma} \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}, \quad S_t = \frac{C_t - H_t}{C_t}$
4	Internal habit (difference)	(Constantinides, 1990)	$M_{t+1} = \delta \frac{(C_{t+1} - bC_t)^{-\gamma}}{(C_t - bC_{t-1})^{-\gamma}}$
5	Internal habit (ratio)	(Abel, 1990)	$M_{t+1} = \delta \frac{(C_{t+1}/C_t^b)^{-\gamma}}{(C_t/C_{t-1}^b)^{-\gamma}}$
6	Long-run risk	(Bansal and Yaron, 2004)	M_{t+1} under Epstein–Zin with $\Delta c_{t+1} = \mu + x_t + \sigma_t \eta_{t+1}$; x_t persistent, σ_t^2 stochastic
7	Long-run risk with jumps	(Drechsler and Yaron, 2011)	Adds compound Poisson jumps in σ_t^2 to the Bansal–Yaron SDF
8	Rare disaster (constant intensity)	(Barro, 2006; Gabaix, 2012)	M_{t+1} includes disaster probability p and contraction size b: $M_{t+1} \propto (1 - p + p \cdot b^{-\gamma})$
9	Variable rare disaster	(Wachter, 2013)	Time-varying disaster intensity λ_t following a Cox process; M_{t+1} depends on λ_t, λ_{t+1}
10	Precautionary savings (prudence)	(Kimball, 1990; Leland, 1968)	$M_{t+1} \approx \delta (C_{t+1}/C_t)^{-\gamma} [1 + \frac{\gamma+1}{2} \sigma_{c,t}^2]$; prudence terms
11	Durable consumption	(Yogo, 2006)	Epstein–Zin with nondurable and durable consumption; M_{t+1} depends on (C_{t+1}/C_t), (D_{t+1}/D_t), and $R_{m,t+1}$
12	Heterogeneous agents	(Constantinides and Duffie, 1996)	$M_{t+1} = \delta (C_{t+1}/C_t)^{-\gamma} \exp[-\frac{\gamma(\gamma+1)}{2} \sigma_{\text{cross}}^2]$; σ_{cross}^2 = cross-sectional cons. variance
13	Nonparametric Euler	(Escanciano et al., 2016)	$M_{t+1} = g(C_{t+1}/C_t)$ with g unknown, identified nonparametrically from Euler equation

The thirteen consumption-based restrictions span several distinct economic channels. Restrictions 1–5 vary preferences while maintaining i.i.d. or simple Markov consumption dy-

namics; what differs is how marginal utility depends on consumption growth and, in the habit models, on the history of consumption. Power utility (Restriction 1) serves as the baseline, and each subsequent preference specification adds a mechanism—intertemporal substitution separated from risk aversion (Restriction 2), slow-moving external habit (Restriction 3), or direct dependence on lagged consumption (Restrictions 4–5). These five SDFs share the same data requirements (aggregate consumption growth) but impose different functional relationships between M_{t+1} and Δc_{t+1}.

Restrictions 6–9 modify the consumption *process* rather than preferences alone. Long-run risk (Restriction 6) introduces a persistent expected-growth component x_t and stochastic volatility σ_t, both of which enter the SDF through Epstein–Zin recursion; the SDF depends on the news to x_t and σ_t^2, not merely on realized consumption growth. Adding jumps (Restriction 7) generates occasional large movements in volatility that are absent under Gaussian dynamics. Rare disaster models (Restrictions 8–9) replace continuous volatility variation with low-probability, high-impact consumption contractions. The economic content differs sharply: long-run risk explains the equity premium through intertemporal hedging demand, while disaster models explain it through the pricing of tail events.

Restrictions 10–13 extend the framework along further dimensions. Precautionary savings (Restriction 10) introduces higher-order risk terms via the prudence coefficient, generating SDF dependence on conditional consumption variance. Durable consumption (Restriction 11) adds a second consumption good with its own intratemporal elasticity. Heterogeneous agents (Restriction 12) replaces the representative consumer with a continuum of agents whose idiosyncratic consumption risk enters the aggregate SDF through the cross-sectional variance of consumption growth. The nonparametric Euler restriction (Restriction 13) abandons parametric preferences entirely, requiring only that some monotone decreasing function of consumption growth prices assets correctly—it serves as a benchmark that detects whether *any* consumption-based SDF is empirically useful, without committing to a specific functional form.

3.2 Production-Based and Investment Restrictions

Production-based asset-pricing models derive expected returns from firms' intertemporal investment decisions. The key insight is that firms equate the marginal cost of investment today to the expected discounted marginal benefit, and this first-order condition links expected stock returns to observable investment and profitability variables. All ten restrictions in this family are Type B (production FOC) penalties as defined in Section 2.3.2:

$$C_j^{\text{prod}}(f) \;=\; \frac{1}{n}\sum_{s=1}^{n}\big(f(\mathbf{X}_s) - g_j(\mathbf{Z}_s^{(j)})\big)^2, \tag{26}$$

where g_j maps firm-level investment characteristics $\mathbf{Z}_{i,t}^{(j)}$ to expected returns. The functional form of g_j and the variables that enter $\mathbf{Z}_{i,t}^{(j)}$ vary across restrictions.

Table 3: Production-Based and Investment Restrictions

#	Restriction	Source	Key Penalty Form
14	Investment CAPM	(Cochrane, 1991; Liu et al., 2009)	$g_{14}(I/K, \text{ROE})$: expected return equals marginal cost of investment via q-theory SDF
15	q-theory factor model	(Hou et al., 2015)	$g_{15}(I/K, \text{ROE}, \text{EG})$: three-factor pricing from investment, profitability, expected growth
16	Convex adjustment costs	(Hayashi, 1982)	$g_{16}(I/K) = \phi'(I_{i,t}/K_{i,t})$: marginal Tobin's q equals marginal adjustment cost
17	Nonconvex adjustment costs	(Abel and Eberly, 1996; Cooper and Haltiwanger, 2006)	$g_{17}(I/K, \mathbb{1}_{\{\lvert I/K\rvert > h\}})$: adds fixed costs and partial irreversibility via indicator for large adjustments
18	Irreversibility	(Abel and Eberly, 1996; Zhang, 2005)	$g_{18}(I/K, \mathbb{1}_{\{I<0\}})$: asymmetric adjustment costs with higher cost of disinvestment
19	Financial constraints (collateral)	(Kiyotaki and Moore, 1997)	$g_{19}(I/K, K/\text{debt})$: expected return premium reflects binding collateral constraint
20	Financial constraints (equity issuance)	(Hennessy and Whited, 2005)	$g_{20}(I/K, \text{leverage}, \text{cash flow})$: equity issuance costs create a wedge in the investment FOC
21	Costly external finance	(Livdan et al., 2009)	$g_{21}(I/K, \text{leverage}, \text{dividends})$: external financing premium enters marginal cost of capital
22	Labor-augmented production	(Belo et al., 2014)	$g_{22}(I/K, \text{HN})$: hiring rate HN enters alongside physical investment via labor adjustment costs
23	Intangible capital	(Eisfeldt and Papanikolaou, 2013)	$g_{23}(I/K, \text{R\&D}/K, \text{SGA}/K)$: intangible investment (R&D, organizational capital) augments physical I/K

The production-based restrictions divide into three groups. Restrictions 14–18 concern the technology of physical capital adjustment. The investment CAPM (Restriction 14) and q-theory (Restriction 15) provide the broadest frameworks: expected returns are determined by firms' optimal investment behavior, with investment-to-capital ratio and profitability as

the key state variables. Restrictions 16–18 refine the adjustment cost specification. Convex adjustment costs (Restriction 16) yield a smooth, monotone relationship between I/K and expected returns. Nonconvex adjustment costs (Restriction 17) and irreversibility (Restriction 18) break this smoothness: large adjustments incur fixed costs, and disinvestment is more expensive than investment. These restrictions generate different functional forms for g_j—Restriction 16 predicts a linear or quadratic relationship in I/K, while Restrictions 17–18 predict threshold effects and asymmetries.

Restrictions 19–21 address financial frictions. These models predict that expected returns reflect not only the marginal product of capital but also the shadow cost of financing constraints. The three restrictions differ in which friction binds: collateral constraints (Restriction 19) link the premium to the ratio of capital to debt, equity issuance costs (Restriction 20) link it to leverage and internal cash flow, and costly external finance (Restriction 21) links it to the dividend payout and leverage. All three predict that financially constrained firms earn higher expected returns, but the mapping from observables to the constraint's shadow value differs across models.

Restrictions 22–23 expand the set of productive inputs beyond physical capital. Labor adjustment costs (Restriction 22) introduce the hiring rate as an additional state variable in the investment FOC, generating a two-dimensional mapping $(I/K, \mathrm{HN}) \mapsto \mathbb{E}[R]$. Intangible capital (Restriction 23) adds R&D spending and selling, general, and administrative expenses as measures of intangible investment, recognizing that knowledge capital and organizational capital are productive assets with their own adjustment costs.

3.3 Intermediary and Institutional Restrictions

Intermediary asset-pricing models posit that marginal investors are not households but financial intermediaries—broker-dealers, banks, hedge funds—whose pricing kernels reflect institutional constraints on leverage, capital, and funding. When intermediaries are the marginal pricers, expected returns depend on intermediary-specific state variables: capital

ratios, leverage, funding spreads, and balance-sheet capacity. All eight restrictions in this family are Type C (factor structure) penalties as defined in Section 2.3.3:

$$C_j^{\text{factor}}(f) \;=\; \frac{1}{n}\sum_{i=1}^{N}\sum_{t=1}^{T}\big(f(\mathbf{X}_{i,t}) - \hat{\beta}_{i,j}\,F_{j,t} - \hat{\alpha}_{i,j}\big)^2, \tag{27}$$

where $F_{j,t}$ is the intermediary factor and $\hat{\beta}_{i,j}$ is asset i's loading on that factor.

Table 4: Intermediary and Institutional Restrictions

#	Restriction	Source	Key Penalty Form
24	Intermediary capital ratio	(He and Krishnamurthy, 2013)	$F_{24,t}$ = intermediary capital ratio; SDF linear in $F_{24,t}$ with time-varying risk prices
25	Intermediary leverage	(Adrian et al., 2014)	$F_{25,t}$ = broker-dealer leverage factor; innovations in leverage price cross-section
26	Funding liquidity	(Brunnermeier and Pedersen, 2009)	$F_{26,t}$ = funding liquidity measure (TED spread, Libor-OIS); margins amplify funding shocks
27	Dealer balance sheet	(He et al., 2017)	$F_{27,t}$ = primary dealer equity capital ratio; single-factor model for broad cross-section
28	Limits to arbitrage / noise trader	(De Long et al., 1990)	$F_{28,t}$ = noise-trader sentiment proxy; mispricing risk is priced because arbitrage is limited
29	Slow-moving capital	(Duffie, 2010)	$F_{29,t}$ = capital flow proxy; price deviations from fundamentals persist due to slow reallocation
30	Margin CAPM	(Gârleanu and Pedersen, 2011)	$F_{30,t}$ = margin requirement factor; expected returns depend on both market beta and margin beta
31	Leverage constraints / BAB	(Frazzini and Pedersen, 2014)	$F_{31,t}$ = betting-against-beta factor; leverage-constrained investors tilt toward high-beta, compressing alpha

The intermediary restrictions separate into models of intermediary health (Restrictions 24–25, 27) and models of market frictions that constrain intermediation (Restrictions 26, 28–31). The intermediary capital ratio (Restriction 24) and leverage (Restriction 25) factors capture the state of broker-dealer balance sheets: when capital is depleted or leverage is high, the

marginal value of wealth to the intermediary rises, increasing risk prices. The dealer balance sheet factor (Restriction 27) constructs an equity capital ratio specifically for primary dealers, providing a more direct measure of the capacity constraint. These three restrictions use different empirical proxies for the same conceptual channel—intermediary distress—and their relative μ_j values reveal which proxy captures the pricing-relevant variation.

Restrictions 26 and 28–31 focus on frictions in the trading environment. Funding liquidity (Restriction 26) links expected returns to the cost and availability of short-term funding: when funding dries up, margins increase and leveraged positions are unwound, creating fire-sale dynamics. Limits to arbitrage (Restriction 28) and slow-moving capital (Restriction 29) introduce a role for mispricing persistence—arbitrageurs face constraints (capital, risk limits, organizational frictions) that prevent immediate price correction. The margin CAPM (Restriction 30) derives a two-beta model in which both market exposure and margin exposure are priced, while the leverage-constraint model (Restriction 31) predicts a compressed security market line because leverage-constrained investors overweight high-beta assets. The key distinction: Restrictions 24–27 model *who* the marginal pricer is, while Restrictions 28–31 model *what constrains* the marginal pricer.

3.4 Information and Learning Restrictions

Information and learning models generate expected-return variation from agents' incomplete knowledge about economic fundamentals. Unlike consumption-based models, where the SDF is pinned down by preferences and the consumption process, these models derive pricing implications from the structure of beliefs—how agents learn, what they are uncertain about, and how they process information. These restrictions generate Type A penalties when they specify a complete SDF (Restrictions 32–35) or Type C penalties when they imply a factor structure driven by a belief-related state variable (Restrictions 36–37).

Table 5: Information and Learning Restrictions

#	Restriction	Source	Key Penalty Form
32	Bayesian learning about growth	(Pástor and Veronesi, 2009)	Type A: M_{t+1} depends on posterior mean and variance of long-run growth rate
33	Learning about regime	(David, 2008; Veronesi, 1999)	Type A: M_{t+1} depends on regime probability $\pi_t = P(\text{good state} \mid \mathcal{F}_t)$; belief updates amplify volatility
34	Parameter uncertainty	(Johannes et al., 2014)	Type A: M_{t+1} integrates over posterior distribution of model parameters; predictive variance enters pricing
35	Ambiguity aversion	(Epstein and Schneider, 2008; Ilut and Schneider, 2014)	Type A: agent evaluates returns under worst-case prior from ambiguity set $\mathcal{P}$; M_{t+1} = worst-case density ratio
36	Disagreement	(Basak, 2005; Xiong and Yan, 2010)	Type C: $F_{36,t}$ = dispersion of beliefs; disagreement generates a priced factor via speculative trading
37	Rational inattention	(Maćkowiak and Wiederholt, 2009; Van Nieuwerburgh and Veldkamp, 2010)	Type C: $F_{37,t}$ = information flow proxy; limited attention capacity generates endogenous information asymmetry

The six information restrictions differ in *what* agents are uncertain about and *how* they process that uncertainty. Bayesian learning (Restriction 32) generates time-varying risk premia because the posterior variance of growth declines as data accumulate, changing the conditional distribution of the SDF. Regime learning (Restriction 33) produces a distinct mechanism: agents infer a latent state (expansion vs. recession), and uncertainty about the current regime amplifies return volatility and risk premia—the SDF depends on the belief π_t rather than on a parameter estimate. Parameter uncertainty (Restriction 34) prices the risk that the true data-generating process is unknown, generating a premium for model risk that shrinks only slowly with sample size.

Ambiguity aversion (Restriction 35) departs from Bayesian updating altogether: the agent entertains a set of plausible models $\mathcal{P}$ and evaluates each asset under the worst-case model. The SDF is the likelihood ratio between the worst-case and reference models, generating pricing implications distinct from those of any single Bayesian model. Disagreement (Restriction 36) and rational inattention (Restriction 37) shift focus from a single agent's beliefs to the cross-section of beliefs. Disagreement models price the dispersion of beliefs as a risk factor (optimists hold long positions, pessimists short, and their trading generates a risk premium). Rational inattention models generate endogenous information asymmetry: agents with limited processing capacity allocate attention optimally, and the resulting heterogeneity in information sets produces cross-sectional variation in expected returns.

3.5 Demand-System and Market Microstructure Restrictions

Demand-system models derive asset prices from the aggregation of heterogeneous investors' portfolio demands. The equilibrium restriction is market clearing: aggregate demand for each asset must equal its supply. This yields Type D (demand-system equilibrium) penalties as defined in Section 2.3.4:

$$C_j^{\text{demand}}(f) \;=\; \sum_{t=1}^{T} \left\| \sum_i w_i^{(j)}\, D_i^{(j)}\big(\hat{\mathbf{f}}_t\big) - \mathbf{S}_t \right\|^2 , \tag{28}$$

where $\hat{\mathbf{f}}_t = (f(\mathbf{X}_{1,t}), \ldots, f(\mathbf{X}_{N,t}))^\top$ and $\mathbf{S}_t$ denotes market-capitalization shares. Restrictions 41–43 involve microstructure frictions and take modified forms as described below.

Table 6: Demand-System and Market Microstructure Restrictions

#	Restriction	Source	Key Penalty Form
38	Market clearing / demand system	(Koijen and Yogo, 2019)	Type D: D_i depends on characteristics and latent demand; aggregate demand = supply at $\hat{\mathbf{f}}_t$
39	Portfolio choice FOC / asset demand	(Koijen et al., 2024)	Type D: D_i derived from first-order conditions with characteristic-based demand elasticities
40	Inelastic markets	(Gabaix and Koijen, 2021)	Type D: aggregate demand inelasticity ϵ amplifies flow shocks; price impact = $\Delta\text{flow}/\epsilon$
41	Search frictions OTC	(Duffie et al., 2005)	Modified Type B: g_{41} includes search intensity and bargaining power; illiquidity premium = $f_{\text{search}}(\lambda, \beta)$
42	Fire sale externalities	(Shleifer and Vishny, 1992)	Modified Type C: $F_{42,t}$ = fire-sale pressure proxy; forced selling by distressed funds depresses prices
43	Short-sale constraints	(Miller, 1977; Hong and Stein, 2003)	Modified Type B: g_{43} predicts overpricing as a function of short interest and disagreement among investors

The demand-system restrictions (38–40) share the market-clearing equilibrium condition but differ in the structure of investors' demand functions. Restriction 38 models demand as a function of observable characteristics and a latent demand shock, estimating the demand system directly from portfolio holdings data. Restriction 39 derives demand from investors' optimality conditions with characteristic-based demand elasticities, providing a micro-founded demand function. Restriction 40 focuses on the *aggregate* demand elasticity: when total market demand is inelastic, flow-driven trading has outsized price impact, and expected returns reflect the accommodation of demand shocks rather than fundamental risk.

Restrictions 41–43 address frictions that prevent instantaneous and costless trading. Search frictions (Restriction 41) apply primarily to over-the-counter markets, where finding a counterparty takes time; the expected-return penalty includes the search intensity and the buyer's bargaining power. Fire sale externalities (Restriction 42) generate a factor-like penalty: when multiple leveraged institutions simultaneously liquidate positions, prices

deviate from fundamentals by an amount proportional to the aggregate selling pressure. Short-sale constraints (Restriction 43) restrict the adjustment mechanism on the negative side: when pessimists cannot short, asset prices reflect only optimists' valuations, and the degree of overpricing depends on the divergence of opinions and the binding intensity of the constraint.

3.6 Variance and Volatility Restrictions

Variance and volatility models link expected returns to measures of conditional variance and the compensation investors demand for bearing variance risk. These two restrictions use publicly available data—the CBOE Volatility Index (VIX) as a measure of implied variance and daily CRSP returns to compute realized variance—rather than individual equity option data, making them applicable to the broad cross-section over the post-1990 period. Both restrictions generate Type B penalties that map volatility-related state variables to expected returns.

Table 7: Variance and Volatility Restrictions

#	Restriction	Source	Key Penalty Form
44	Variance risk premium	(Bollerslev et al., 2009)	Type B: $g_{44}(\text{VRP}_t) = \text{IV}_t^2 - \mathbb{E}_t[\text{RV}_{t+1}]$; VRP predicts aggregate returns
45	Stochastic volatility	(Heston, 1993)	Type B: $g_{45}(V_t)$; conditional variance V_t follows CIR process; variance risk carries its own risk premium

The two variance and volatility restrictions capture complementary aspects of volatility pricing. The variance risk premium (Restriction 44) uses the spread between implied variance (from VIX) and expected realized variance (computed from daily CRSP returns within each month) as a predictor of aggregate returns; the penalty penalizes discrepancies between $f(\mathbf{X}_t)$ and the VRP-implied expected return. Bollerslev et al. (2009) show that VRP is a strong predictor of market returns at horizons from one to six months, and the

restriction encodes this predictive relationship. Stochastic volatility (Restriction 45) uses the conditional variance V_t from a Heston-type model, where VIX serves as a proxy for the latent conditional variance and the variance process carries its own risk premium distinct from the equity premium. Together, these restrictions ensure that the estimator respects the well-documented relationship between variance dynamics and expected returns without requiring granular option-level data.

3.7 Behavioral and Sentiment Restrictions

Behavioral models derive expected-return patterns from departures from rational expectations or expected utility. These restrictions do not derive an SDF from a representative agent's Euler equation; instead, they predict cross-sectional or time-series return patterns as functions of behavioral variables—past returns, capital gains, sentiment indices. Most generate Type B penalties, where g_j maps behavioral state variables to expected returns.

Table 8: Behavioral and Sentiment Restrictions

#	Restriction	Source	Key Penalty Form
46	Prospect theory / loss aversion	(Barberis et al., 2001)	Type A: M_{t+1} modified by loss aversion coefficient λ around reference point; kink at zero gain
47	Overconfidence	(Daniel et al., 1998)	Type B: $g_{47}(\text{momentum}, \text{reversal})$; overconfident agents generate short-run continuation, long-run reversal
48	Disposition effect / capital gains overhang	(Frazzini, 2006)	Type B: $g_{48}(\text{CGO}_{i,t})$; capital gains overhang = unrealized gain/loss relative to reference price
49	Extrapolative expectations	(Barberis et al., 2018)	Type B: $g_{49}(\bar{R}_{i,t}^{\text{past}})$; agents extrapolate recent returns, generating momentum and mean reversion
50	Investor sentiment	(Baker and Wurgler, 2006)	Type C: $F_{50,t}$ = Baker–Wurgler sentiment index; high sentiment predicts low returns for speculative stocks
51	Salience theory	(Bordalo et al., 2012)	Type B: $g_{51}(\text{salience}_{i,t})$; agents overweight salient (extreme) payoffs, distorting subjective probabilities

The behavioral restrictions exploit distinct psychological mechanisms. Prospect theory (Restriction 46) modifies the SDF with a kink at the reference point: losses are weighted more heavily than gains by a factor $\lambda > 1$, generating a first-order risk premium for assets whose returns covary with movements around the reference point. This is the only behavioral restriction that directly specifies an SDF, placing it closest to the consumption-based framework.

Restrictions 47–49 generate cross-sectional return predictability from belief formation biases. Overconfidence (Restriction 47) produces momentum at short horizons (agents overreact to private signals, pushing prices beyond fundamentals) and reversal at long horizons (prices revert as public information arrives). The disposition effect (Restriction 48) predicts return continuation among stocks with large unrealized gains and return reversal among stocks with large unrealized losses, because investors' reluctance to realize losses impedes price adjustment. Extrapolative expectations (Restriction 49) model agents who weight recent returns too heavily when forming expectations, generating a specific functional form for the relationship between past returns and expected future returns. These three restrictions share momentum-related variables as inputs but predict different functional relationships between past returns and expected future returns.

Investor sentiment (Restriction 50) operates at the aggregate level: the Baker–Wurgler sentiment index predicts that, conditional on high sentiment, speculative stocks (small, unprofitable, high-volatility) earn low subsequent returns, while safe stocks are largely unaffected. This generates a Type C penalty where $F_{50,t}$ is the sentiment index and the loadings $\hat{\beta}_{i,50}$ are heterogeneous across stocks, depending on firm characteristics. Salience theory (Restriction 51) predicts mispricing at the individual asset level: stocks with salient (extreme) upside payoffs are overpriced because agents overweight the probability of salient outcomes, generating a negative relationship between salience and expected returns.

3.8 Macro-Finance and Term Structure Restrictions

Macro-finance models link expected returns on equities and bonds to macroeconomic state variables: the yield curve, inflation expectations, monetary policy, and consumption-wealth ratios. These restrictions enter as Type B or Type C penalties depending on whether the model implies a direct functional mapping from macro variables to expected returns (Type B) or a factor structure driven by a macro factor (Type C).

Table 9: Macro-Finance and Term Structure Restrictions

#	Restriction	Source	Key Penalty Form
52	Affine term structure no-arbitrage	(Duffie and Kan, 1996; Ang and Piazzesi, 2003)	Type B: $g_{52}(\mathbf{y}_t)$; no-arbitrage restrictions on yield curve factors constrain equity risk premia via the state vector $\mathbf{y}_t$
53	Inflation risk premium	(Bekaert and Engstrom, 2010)	Type C: $F_{53,t}$ = inflation risk premium from nominal-real yield spread; inflation uncertainty priced in equities
54	Monetary policy / Taylor rule	(Bernanke and Kuttner, 2005)	Type B: $g_{54}(\Delta i_t^{\text{surprise}})$; surprise component of federal funds rate changes predicts equity returns
55	Consumption-wealth ratio (cay)	(Lettau and Ludvigson, 2001)	Type B: $g_{55}(cay_t)$; cointegrating residual of consumption, asset wealth, and labor income predicts returns
56	Credit spread / EBP	(Gilchrist and Zakrajšek, 2012)	Type C: $F_{56,t}$ = excess bond premium; credit market distress predicts equity returns via risk appetite channel

The macro-finance restrictions link equity expected returns to macroeconomic and fixed-income state variables through five distinct channels. The affine term structure restriction (Restriction 52) imposes no-arbitrage across maturities, using the yield curve state vector $\mathbf{y}_t$ (level, slope, curvature, and potentially macro factors) to constrain the time-varying equity risk premium. This restriction is structurally different from the others because it derives from the bond market and imposes cross-market no-arbitrage.

Restrictions 53–54 transmit monetary and inflation shocks to equity prices. The inflation risk premium (Restriction 53) uses the spread between nominal and real yields to extract

the compensation investors demand for bearing inflation uncertainty; this premium predicts equity returns because inflation uncertainty covaries with real economic risk. The monetary policy restriction (Restriction 54) identifies the surprise component of federal funds rate decisions (using fed funds futures) and maps it to equity returns, exploiting the direct channel from policy rates to discount rates.

Restrictions 55 and 56 use slow-moving macroeconomic aggregates. The consumption-wealth ratio *cay* (Restriction 55) exploits cointegration among consumption, asset wealth, and labor income: when consumption is high relative to wealth, expected returns are high because the ratio mean-reverts through subsequent returns. The excess bond premium (Restriction 56) isolates the component of credit spreads that reflects risk appetite and financial intermediary health, net of default risk; it predicts equity returns because deteriorating credit conditions signal tightening financial constraints economy-wide.

3.9 Summary and Discussion

The catalog comprises fifty-six restrictions organized into eight families. Each restriction imposes a distinct functional form on the mapping from observable characteristics $\mathbf{X}_{i,t}$ to expected returns $f(\mathbf{X}_{i,t})$, and each translates into a specific penalty functional $C_j(f)$ within the Theory-KRR objective (2). The penalties are heterogeneous in structure: Type A penalties enforce Euler equation moment conditions and require specifying the SDF (Restrictions 1–13, 32–35, 46); Type B penalties equate expected returns to functions of investment or macro variables through first-order conditions (Restrictions 14–23, 41, 43, 44–45, 47–49, 51–52, 54–55); Type C penalties impose factor-structure restrictions driven by intermediary, sentiment, or macro factors (Restrictions 24–31, 36–37, 42, 50, 53, 56); and Type D penalties enforce market-clearing equilibrium conditions from demand-system models (Restrictions 38–40). Despite this heterogeneity, the framework in Section 2 accommodates all fifty-six penalties within a single objective: each $C_j(f)$ depends on f only through its evaluations at observed data points, the representer theorem applies, and cross-validation selects the vector $\boldsymbol{\mu}$ that

balances predictive fit against theoretical guidance.

A property of the catalog that makes the μ_j ranking meaningful is that many restrictions share observable input variables but impose different functional relationships. Consumption growth enters Restrictions 1–13 through thirteen different SDFs. Investment-to-capital ratio enters Restrictions 14–18 through five different adjustment cost specifications. Past returns enter Restrictions 47–49 through three different behavioral mechanisms. When two restrictions share inputs but one receives a higher μ_j in cross-validation, the data have identified which functional relationship between those inputs and expected returns is more empirically useful. This is the sense in which the cross-validated multipliers $\boldsymbol{\mu}$ produce a ranking of economic theories: they rank not just which state variables matter, but which *economic mechanisms linking state variables to expected returns* are supported by the data.

4 Estimation and Implementation

This section addresses the practical aspects of taking Theory-KRR from the mathematical framework of Sections 2–3 to an implementable estimator. We describe the kernel choice and bandwidth calibration (Section 4.1), the pre-estimation of structural parameters that enter the penalty functionals (Section 4.2), the cross-validation strategy for selecting the high-dimensional vector of penalty multipliers (Section 4.3), computational details (Section 4.4), and the rolling-window procedure for out-of-sample evaluation (Section 4.5).

4.1 Kernel Choice and Bandwidth Selection

The kernel $k : \mathbb{R}^p \times \mathbb{R}^p \to \mathbb{R}$ determines the RKHS $\mathcal{H}$ in which the estimator searches for the conditional expectation function f^*. Our primary specification is the Gaussian radial basis function (RBF) kernel:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right), \tag{29}$$

where $\sigma > 0$ is the bandwidth parameter and $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^p$. The Gaussian kernel is universal: the associated RKHS is dense in the space of continuous functions on any compact subset of $\mathbb{R}^p$, so the approximation error $\inf_{f\in\mathcal{H}} \|f - f^*\|$ can be made arbitrarily small. This universality is essential for our application, where the true conditional expectation function may exhibit the nonlinearities, interactions, and threshold effects documented in the cross-section of returns.

We set the bandwidth σ using the median heuristic:

$$\sigma = \text{median}\{\|\mathbf{X}_{i,t} - \mathbf{X}_{j,s}\| : (i,t) \neq (j,s)\}\,. \tag{30}$$

The median heuristic places the kernel bandwidth at a scale where roughly half of the pairwise distances produce kernel evaluations above $e^{-1/2} \approx 0.61$ and half below. This data-driven rule avoids both extremes: a bandwidth that is too small relative to the data's spread produces a nearly diagonal kernel matrix (overfitting), while a bandwidth that is too large produces a nearly constant kernel matrix (underfitting, approaching the global mean). The median heuristic has strong theoretical support in the kernel methods literature and performs well in high-dimensional settings where manual tuning is infeasible.

As a robustness check, we also consider the Matérn-5/2 kernel:

$$k_{5/2}(\mathbf{x},\mathbf{x}') = \left(1 + \frac{\sqrt{5}\,r}{\ell} + \frac{5\,r^2}{3\,\ell^2}\right)\exp\left(-\frac{\sqrt{5}\,r}{\ell}\right), \qquad r = \|\mathbf{x} - \mathbf{x}'\|, \tag{31}$$

where $\ell > 0$ is the length-scale parameter. Functions in the Matérn-5/2 RKHS are twice but not three times differentiable, producing smoother fitted surfaces than the infinitely differentiable Gaussian kernel. If the true f^* has limited smoothness, the Matérn kernel may provide a better bias-variance trade-off.

Two limiting cases clarify the role of the kernel. A linear kernel $k(\mathbf{x},\mathbf{x}') = \mathbf{x}^\top\mathbf{x}'$ restricts $\mathcal{H}$ to linear functions $f(\mathbf{x}) = \boldsymbol{\beta}^\top\mathbf{x}$, and Theory-KRR reduces to a penalized linear regression with structural penalties on coefficients. This nesting (Proposition 19(iii)) allows us to

isolate the contribution of nonlinearity from the contribution of theory-informed penalties. At the other extreme, the Gaussian kernel with bandwidth $\sigma \to 0$ produces a kernel matrix approaching the identity, so $\hat{f}$ interpolates the data subject only to the structural penalties. The joint configuration of σ, λ_{stat}, and $\boldsymbol{\mu}$ determines the estimator's regularization, and cross-validation selects the balance that minimizes OOS prediction error.

4.2 Pre-Estimation of Structural Parameters

Each structural penalty $C_j(f)$ depends on model-specific parameters θ_j that characterize the underlying economic model. For consumption-based models, θ_j includes the coefficient of relative risk aversion γ, the elasticity of intertemporal substitution ψ, the subjective discount factor δ, and the habit persistence parameter b. For production-based models, θ_j governs the curvature of the adjustment cost function. For intermediary models, θ_j parameterizes the leverage constraint or capital requirement.

We treat these structural parameters as known at estimation time, obtained in a first stage that is separate from the KRR optimization. Two approaches are available.

Calibration from the literature. For parameters with well-established values, we adopt published calibrations from the original papers. We set the subjective discount factor $\delta = 0.99$ (monthly) in all consumption-based models and use the habit persistence parameter $b = 0.87$ from Campbell and Cochrane (1999) for the external habit model. Transparency and reproducibility are the primary advantages: the penalty $C_j(f)$ is fully determined by the data and a known parameter vector, with no researcher discretion at this stage. The risk is that published parameter values, typically calibrated to match aggregate moments in a specific sample, may not be appropriate for our cross-sectional return prediction task.

Pre-estimation from macro data. For parameters that are sample-specific or that lack consensus values, we pre-estimate $\hat{\theta}_j$ by GMM applied to aggregate consumption Euler equa-

tions. For each consumption-based SDF model j with SDF $M_j(\theta_j, \mathbf{X}_t, \mathbf{X}_{t+1})$, we estimate

$$\hat{\theta}_j = \arg\min_{\theta_j} \left\| \frac{1}{T} \sum_{t=1}^{T} \mathbf{z}_t \left[M_j(\theta_j, \mathbf{X}_t, \mathbf{X}_{t+1}) \, R^{\text{gross}}_{m,t+1} - 1 \right] \right\|^2_{\hat{\mathbf{W}}^{-1}}, \tag{32}$$

where $\mathbf{z}_t$ is a vector of instruments (lagged consumption growth, the dividend-price ratio, the term spread) and $\hat{\mathbf{W}}$ is an HAC-consistent estimate of the instrument covariance matrix. For production-based models, we estimate adjustment cost parameters from firm-level investment regressions. The pre-estimated $\hat{\theta}_j$ is then plugged into $C_j(f)$ and treated as fixed during the second-stage KRR optimization.

We adopt a hybrid strategy: calibrate parameters for which the literature provides precise, widely accepted values (δ, b), and pre-estimate parameters that are sensitive to the sample period or data frequency (γ, ψ). To assess sensitivity, we report results over a grid of values for the most influential parameters:

$$\gamma \in \{2, 5, 10, 20\}, \qquad \delta \in \{0.95, 0.97, 0.99\}. \tag{33}$$

This grid spans the range of calibrations used in the asset-pricing literature, from low-risk-aversion economies ($\gamma = 2$) to the high-risk-aversion calibrations required by rare disaster models ($\gamma = 20$).

4.3 Cross-Validation Strategy

The full hyperparameter vector of Theory-KRR is $(\lambda_{\text{stat}}, \mu_1, \ldots, \mu_J)$, where $J = 56$ in our baseline specification. A 57-dimensional grid search is computationally infeasible. We reduce dimensionality by exploiting the economic structure of the restrictions.

Grouped penalties. We group the 56 restrictions into $G = 8$ families corresponding to the eight economic model classes cataloged in Section 3: consumption-based Euler equations, production-based restrictions, intermediary restrictions, information and learning restric-

tions, demand-system restrictions, variance and volatility restrictions, behavioral restrictions, and macro-finance restrictions. Within each family g, we impose a common multiplier:

$$\mu_j = \mu_{g(j)}^{\text{group}} \qquad \text{for all } j \text{ in family } g. \tag{34}$$

This reduces the search space from 61 dimensions to 9: one statistical penalty λ_{stat} and eight group-level multipliers $\mu_1^{\text{group}}, \ldots, \mu_8^{\text{group}}$. The grouping is motivated by the observation that restrictions within the same economic family share a common source of identification and are likely to have correlated empirical content. The grouping is not restrictive in the dimension that matters: cross-validation can set any group's multiplier to zero, shutting down that entire family of restrictions.

Hierarchical penalties. Within each group, individual restrictions may differ in empirical tightness. An adaptive refinement assigns restriction-specific weights:

$$\mu_j = \mu_{g(j)}^{\text{group}} \cdot w_j, \qquad w_j = \frac{1}{\tilde{C}_j(\hat{f}^{\text{KRR}}) + \epsilon}, \tag{35}$$

where $\hat{f}^{\text{KRR}}$ is the unrestricted (standard) KRR estimator and $\epsilon > 0$ is a small constant that prevents division by zero. The weight w_j is inversely proportional to the empirical violation of restriction j under the unrestricted estimator. Restrictions that the data nearly satisfy on their own receive large weights (strong enforcement), while restrictions that the data grossly violate receive small weights (weak enforcement). This mirrors the adaptive penalty logic of the LLM-Lasso, where text-derived priors modulate variable-specific penalty weights: here, the unrestricted fit serves as the prior on which restrictions are empirically plausible.

Temporal cross-validation. Time-series dependence in asset returns invalidates standard K-fold cross-validation. We use leave-one-year-out (LOYO) temporal CV. For each year s in the evaluation period, we train on all data excluding year s and predict returns in year s. The

CV criterion selects the hyperparameter vector that minimizes average squared prediction error across left-out years:

$$(\hat{\lambda}_{\text{stat}}, \hat{\boldsymbol{\mu}}^{\text{group}}) = \underset{\lambda_{\text{stat}}, \boldsymbol{\mu}^{\text{group}}}{\arg\min} \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \frac{1}{n_s} \sum_{(i,t) \in s} \left(R_{i,t+1} - \hat{f}_{-s}(\mathbf{X}_{i,t}) \right)^2, \tag{36}$$

where $\hat{f}_{-s}$ denotes the estimator trained on all data excluding year s, and n_s is the number of observations in year s. LOYO preserves temporal separation between training and evaluation sets, preventing look-ahead bias.

Search strategy. We search the 9-dimensional hyperparameter space using random search with 200 draws. Each hyperparameter is drawn independently from a log-uniform distribution on $[10^{-4}, 10^{2}]$:

$$\log_{10}(\lambda_{\text{stat}}),\ \log_{10}(\mu_g^{\text{group}}) \overset{\text{iid}}{\sim} \text{Uniform}(-4, 2), \qquad g = 1, \ldots, 8. \tag{37}$$

Random search over log-uniform distributions is well suited to hyperparameter spaces where the optimal values may span several orders of magnitude, as is typical in regularization problems. With 200 random draws, the probability of sampling within the top 5% of the hyperparameter space in each of the 9 dimensions exceeds 0.99.

Each draw requires solving one KRR problem by inverting an $n \times n$ linear system. For $n \approx 4{,}000$ (approximately 370 portfolios $\times$ 12 months), each solve takes roughly 30 seconds on a modern workstation. The full random search over 200 draws completes in approximately 100 minutes per rolling window.

4.4 Computational Details

By Proposition 13, the estimator $\hat{f}(\cdot) = \sum_{s=1}^{n} \hat{\alpha}_s \, k(\mathbf{X}_s, \cdot)$ is parameterized by the coefficient vector $\hat{\boldsymbol{\alpha}} \in \mathbb{R}^n$. The optimization problem (9) is therefore finite-dimensional: we optimize over $\boldsymbol{\alpha}$ rather than over functions in $\mathcal{H}$.

Quadratic penalties (Types A, B, C). When all active penalties are quadratic in the fitted values $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$, the objective is a convex quadratic in $\boldsymbol{\alpha}$, and the solution is given by the linear system (17). The dominant computational cost is the $O(n^3)$ matrix inversion (or, equivalently, the Cholesky decomposition of the $n \times n$ system matrix). We solve the system using Cholesky factorization, which is numerically stable and exploits the positive definiteness guaranteed by $\lambda_{\text{stat}} > 0$.

For problems involving Type A (Euler equation) penalties, the system matrix in (16) contains the term $\boldsymbol{\Omega}\,\mathbf{K}$, where $\boldsymbol{\Omega}$ includes the $\mathbf{A}_j^\top \mathbf{A}_j$ matrices. These are at most rank T (since $\mathbf{A}_j$ has T rows), and we exploit this low-rank structure by precomputing and caching the products $\mathbf{A}_j\,\mathbf{K}$ for each active Euler-equation restriction.

Nonquadratic penalties (Type D). When demand-system penalties with nonlinear demand functions are active, the objective is nonconvex. We minimize using L-BFGS-B with analytically computed gradients. The standard KRR solution (all $\mu_j = 0$) provides a warm start that places the optimizer in the basin of attraction of the theory-informed minimum. In practice, we use L-BFGS-B for all penalty types, as the analytical Hessians required by the augmented linear system approach are computationally expensive for the full restriction set. Convergence requires 50–200 L-BFGS iterations, each costing $O(n^2)$ for the gradient computation via analytically derived penalty gradients.

Memory and scalability. The kernel matrix $\mathbf{K}$ is $n \times n$ and must be stored in memory. For our primary specification with $n \approx 4{,}000$ (cross-section of managed portfolios over 12 months), $\mathbf{K}$ occupies approximately 128 MB in double precision, which is manageable on standard hardware. For the full individual-stock panel ($n > 100{,}000$), storing $\mathbf{K}$ requires over 80 GB, which exceeds typical workstation memory.

Two standard approximations address this scaling challenge. The Nyström approximation selects a subset of $m \ll n$ landmark points and approximates $\mathbf{K} \approx \mathbf{K}_{nm}\,\mathbf{K}_{mm}^{-1}\,\mathbf{K}_{nm}^\top$, where $\mathbf{K}_{nm}$ is the $n \times m$ matrix of kernel evaluations between all points and the landmarks,

and $\mathbf{K}_{mm}$ is the $m \times m$ kernel matrix among landmarks. This reduces memory to $O(nm)$ and solve time to $O(nm^2)$. Random Fourier features offer an alternative: they approximate the Gaussian kernel by an explicit feature map $\boldsymbol{\phi} : \mathbb{R}^p \to \mathbb{R}^D$ such that $k(\mathbf{x}, \mathbf{x}') \approx \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{x}')$, converting the kernel method into a linear regression in D-dimensional feature space. Both approximations introduce a controllable approximation error that decreases as m or D increases. We use $m = 500$ Nyström landmarks when the full kernel matrix is infeasible.

Software. The estimator is implemented in Python using NumPy and SciPy for linear algebra. Kernel matrices are constructed via `scipy.spatial.distance`, the Cholesky factorization uses `scipy.linalg.cho_factor`, and L-BFGS-B optimization uses `scipy.optimize.minimize`. Baseline models (OLS, Ridge, Lasso, Elastic Net, Random Forest) use scikit-learn; the neural network baseline uses PyTorch. The full codebase, including cross-validation, rolling-window evaluation, and portfolio construction, is available in the replication package.

4.5 Rolling-Window Procedure

We evaluate Theory-KRR in a rolling out-of-sample framework that mirrors the timing conventions of the asset-pricing literature.

Training window. We use an expanding window with a minimum training length of 240 months (20 years). The first training window spans January 1960 to December 1979; the first OOS prediction is for January 1980. As the evaluation period advances, the training window grows to incorporate all available historical data up to the most recent month. An expanding window maximizes the training sample at each evaluation date and allows the estimator to learn from the full history of cross-sectional return patterns.

Evaluation and rebalancing. At each annual rebalancing date (December of year s), we train Theory-KRR on the full training sample ending in month $s \times 12$ and predict returns for the next 12 months. Annual rebalancing reduces computational cost (one full estimation

per year rather than per month) and aligns with the holding-period assumptions of many characteristic-based strategies. Portfolio weights are set at the rebalancing date and held fixed for 12 months.

Hyperparameter re-estimation. The full hyperparameter vector $(\hat{\lambda}_{\text{stat}}, \hat{\boldsymbol{\mu}}^{\text{group}})$ is re-estimated by LOYO cross-validation at each annual rebalancing date. This allows the optimal balance between statistical and theory-informed regularization to vary over time. Tracking $\hat{\mu}_g^{\text{group}}$ across rebalancing dates reveals temporal variation in the value of each theory family—for instance, whether consumption-based restrictions become more useful during recessions or whether intermediary restrictions gain importance after financial crises. We report the time series of estimated multipliers in Section 6.

Look-ahead bias prevention. All information used in estimation—returns, characteristics, macro state variables, and the structural parameters $\hat{\theta}_j$—is available as of the training window's end date. Characteristics are lagged by at least one month relative to the return being predicted. Macro variables use the vintage available at the prediction date, not subsequently revised values. The structural parameters $\hat{\theta}_j$ are pre-estimated using only data within the training window.

5 Data and Baseline Models

5.1 Data

Returns. We obtain monthly stock returns from the Center for Research in Security Prices (CRSP) monthly stock file. The sample includes ordinary common shares (share codes 10 and 11) listed on NYSE, AMEX, and NASDAQ (exchange codes 1, 2, and 3). Excess returns are computed by subtracting the one-month Treasury bill rate from the Fama-French data library. The effective sample period begins in July 1963—when both CRSP and Compustat

coverage become reliable—and ends in December 2023, spanning 726 months and approximately 18,000 unique securities. This start date matches Gu et al. (2020) and the bulk of the cross-sectional return prediction literature.

Firm characteristics. We use 220 firm-level predictive characteristics constructed from CRSP returns, Compustat fundamentals, and the Open Source Asset Pricing (OSAP) dataset of Jensen et al. (2023). The OSAP dataset provides a standardized, publicly available panel of signed characteristics at the security-month level, constructed following the original papers' methodologies and validated against published results. Supplementary variables include CRSP-derived signals (price, size, short-term reversal) and Compustat-derived fundamentals (investment-to-capital, ROE, leverage, ROA, gross profitability, asset growth, R&D intensity) that enter the structural restriction penalties directly. All characteristics are signed so that higher values predict higher expected returns. This characteristic set subsumes and extends the 94 characteristics used in Gu et al. (2020).

We merge the OSAP characteristics with CRSP returns using the permanent security identifier (permno). To ensure comparability across variables measured in different units, we cross-sectionally rank each characteristic to the unit interval $[0, 1]$ each month. Ranking is a monotone transformation that preserves the ordinal information in each characteristic while eliminating outliers and ensuring that all variables enter the kernel on a common scale.

The structural restrictions in Section 3 additionally require firm-level accounting data from Compustat. We obtain annual fundamentals (525,028 firm-years, 40,678 unique firms, 1950–2025) and quarterly fundamentals (1.87 million firm-quarters, 39,568 unique firms, 1961–2025) from WRDS. These are merged with CRSP via the CRSP-Compustat linking table using primary link types (LC and LU) with an effective link date filter. Key accounting variables include total assets, book equity, sales, income before extraordinary items, capital expenditures, property-plant-equipment, depreciation, long-term and short-term debt, cash holdings, R&D expenditure, and selling-general-administrative expenses. Our Compustat

sample includes both active and inactive firms to avoid survivorship bias.

Macro state variables. We augment firm-level characteristics with macroeconomic state variables that serve two roles: they enter the kernel as predictors of expected returns, and they supply inputs to the structural restriction penalties in Section 3. These variables are drawn from four sources.

From the Federal Reserve Economic Data (FRED), we obtain the 3-month Treasury bill rate (TB3MS), the 10-year Treasury yield (GS10), Moody's AAA and BAA corporate bond yields, the Consumer Price Index (CPIAUCSL), industrial production (INDPRO), and the unemployment rate (UNRATE). The term spread (GS10 minus TB3MS) and the default spread (BAA minus AAA) are computed as differences. From the National Income and Product Accounts via FRED, we obtain personal consumption expenditures on nondurable goods and services, the PCE price index, and population, which we combine to construct real per capita consumption growth—the central input to the thirteen consumption-based Euler equation restrictions.

The intermediary capital ratio and intermediary capital risk factor of He et al. (2017) are obtained from the authors' website at quarterly frequency and interpolated to monthly. The consumption-wealth ratio *cay* of Lettau and Ludvigson (2001) is constructed following their Dynamic OLS methodology from FRED series on household net worth (TNWBSHNO), employee compensation (COE), and consumption, using 8 leads and lags of the differenced regressors. The Baker and Wurgler (2006) investor sentiment index (orthogonalized version) is obtained from the authors' data repository and covers 1965–2023. The CBOE Volatility Index (VIX) is obtained from FRED and is available from January 1990; we aggregate daily values to monthly averages. The excess bond premium of Gilchrist and Zakrajšek (2012) is proxied by the Moody's BAA spread over the 10-year Treasury (BAAFFM series from FRED). Additional macro predictors—the dividend-price ratio, earnings-price ratio, aggregate book-to-market, net equity expansion, Treasury bill rate, long-term yield, and

stock variance—are obtained from the Welch and Goyal (2008) dataset, which covers 1871–2024 at monthly frequency.

Missing data and sample restrictions. We require a minimum of 15 non-missing characteristics per stock-month observation. Observations with fewer non-missing characteristics are dropped. Returns are winsorized at the 0.1st and 99.9th percentiles each month to limit the influence of extreme observations while preserving the bulk of the return distribution. After applying all filters, the sample contains 1,778,850 stock-month observations spanning 18,000 unique securities, covering an average of approximately 2,450 stocks per month.

Not all macro state variables are available for the full sample period. The VIX is available only from 1990, so the two variance and volatility restrictions (44–45) are evaluated on the post-1990 subsample. The HKM intermediary factors are available quarterly from 1970 to 2012; restrictions 24–27 are evaluated on this subsample. The Baker-Wurgler sentiment index is available from 1965 to 2023. All other macro variables cover the full 1963–2023 sample period.

[**Table 1 about here**]

Table 1 reports summary statistics. Panel A presents the distribution of monthly excess returns and selected firm-level characteristics, reporting the mean, standard deviation, and selected percentiles (5th, 25th, 50th, 75th, 95th) across all stock-month observations. Panel B reports summary statistics for the macroeconomic state variables. Panel C documents sample coverage over time, showing the number of stocks per month across the sample period.

5.2 Baseline Models

We compare Theory-KRR against eight baseline models spanning the spectrum from simple linear methods to flexible nonparametric estimators. All baselines use the same data, the same rolling-window procedure described in Section 4.5, and the same LOYO temporal cross-validation for hyperparameter selection. This ensures that performance differences reflect

the models' functional form and regularization, not differences in data or evaluation protocol.

1. **Historical mean.** The simplest benchmark: $\hat{f}(\mathbf{X}_{i,t}) = \bar{R}$, the pooled cross-sectional average of excess returns in the training sample. This is the natural benchmark for R^2_{OOS}, which measures improvement over the historical mean.

2. **OLS.** Linear regression of next-month excess returns on the full vector of characteristics: $\hat{f}(\mathbf{X}_{i,t}) = \hat{\boldsymbol{\beta}}^\top \mathbf{X}_{i,t}$. With 37 characteristics, OLS is prone to overfitting in small training samples but provides a benchmark for the value of nonlinearity.

3. **Ridge regression.** OLS with an ℓ_2 penalty on the coefficient vector: the objective is $\frac{1}{n}\sum_s (R_s - \boldsymbol{\beta}^\top \mathbf{X}_s)^2 + \lambda \|\boldsymbol{\beta}\|_2^2$. Ridge regression shrinks coefficients toward zero without enforcing sparsity. The regularization parameter λ is selected by LOYO CV.

4. **Lasso.** OLS with an ℓ_1 penalty: the objective is $\frac{1}{n}\sum_s (R_s - \boldsymbol{\beta}^\top \mathbf{X}_s)^2 + \lambda \|\boldsymbol{\beta}\|_1$. The ℓ_1 penalty performs variable selection by setting some coefficients exactly to zero.

5. **Elastic net.** A convex combination of ℓ_1 and ℓ_2 penalties: the objective is $\frac{1}{n}\sum_s (R_s - \boldsymbol{\beta}^\top \mathbf{X}_s)^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2$. The elastic net combines variable selection (Lasso) with coefficient shrinkage (ridge). Both λ_1 and λ_2 are selected by LOYO CV.

6. **Standard KRR.** The Gaussian kernel ridge regression estimator with RKHS-norm regularization only: this is Theory-KRR with all $\mu_j = 0$. Standard KRR captures nonlinearities and interactions through the Gaussian kernel but receives no guidance from economic theory. Comparing Theory-KRR to standard KRR isolates the marginal contribution of the structural penalties.

7. **Neural network.** A feedforward neural network with three hidden layers, following the architecture of Gu, Kelly, and Xiu (2020): layer widths of 32, 16, and 8 neurons, ReLU activation functions, batch normalization, and dropout ($p = 0.5$) for regularization. The network is trained by stochastic gradient descent with early stopping based on validation loss (the last 20% of the training window serves as the validation set).

8. **Random forest.** An ensemble of 300 regression trees with the number of candidate variables at each split set to $\lfloor p/3 \rfloor$, following Gu, Kelly, and Xiu (2020). Tree depth is controlled by requiring a minimum of 5 observations per leaf node.

5.3 Evaluation Metrics

We evaluate predictive performance along three dimensions: statistical accuracy, portfolio profitability, and economic value.

Out-of-sample R^2. Following Campbell and Thompson (2008), we measure predictive accuracy by

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{(i,t)\in\mathcal{T}_{\text{OOS}}} \left(R_{i,t+1} - \hat{f}(\mathbf{X}_{i,t})\right)^2}{\sum_{(i,t)\in\mathcal{T}_{\text{OOS}}} \left(R_{i,t+1} - \bar{R}\right)^2}, \tag{38}$$

where $\mathcal{T}_{\text{OOS}}$ indexes the out-of-sample observations and $\bar{R}$ is the historical mean computed from the training sample. A positive R^2_{OOS} indicates that the model predicts better than the historical mean. In the cross-section of individual stock returns, R^2_{OOS} values above 1% are economically meaningful given the high idiosyncratic variance of monthly returns.

Portfolio Sharpe ratios. We form decile portfolios each month by sorting stocks on their predicted expected returns $\hat{f}(\mathbf{X}_{i,t})$. The long-short portfolio goes long the top decile and short the bottom decile, with value-weighted holdings within each decile. We report annualized Sharpe ratios:

$$\text{SR} = \frac{\hat{\mu}_{L-S}}{\hat{\sigma}_{L-S}} \times \sqrt{12}, \tag{39}$$

where $\hat{\mu}_{L-S}$ and $\hat{\sigma}_{L-S}$ are the sample mean and standard deviation of the monthly long-short return. Higher Sharpe ratios indicate that the model's predictions generate more profitable trading strategies per unit of risk.

Certainty equivalent return. For a mean-variance investor with relative risk aversion $\gamma = 5$, the certainty equivalent return of a portfolio with mean $\hat{\mu}_p$ and variance $\hat{\sigma}_p^2$ is

$$\text{CER} = \hat{\mu}_p - \frac{\gamma}{2}\,\hat{\sigma}_p^2. \tag{40}$$

The CER provides an economically interpretable measure of the value of predictability: it is the riskless return that the investor would accept in lieu of the risky portfolio. We compute the CER for the long-short portfolio constructed from each model's predictions.

Statistical inference. We assess the statistical significance of performance differences using three procedures. The Diebold-Mariano test compares squared forecast errors between Theory-KRR and each baseline, with Newey-West standard errors (6 lags) to account for serial correlation. For Sharpe ratio differences, we use a block bootstrap with 12-month blocks and 10,000 replications; the block length preserves the annual dependence structure in monthly returns. All reported standard errors use the Newey-West HAC estimator with 6 lags unless otherwise noted.

5.4 Summary Statistics

[**Table 1 about here**]

Table 1 documents the key features of our dataset. Panel A shows that the average monthly excess return is 0.9% with a standard deviation of 14.2%, consistent with well-known properties of the cross-section. Firm characteristics exhibit substantial cross-sectional dispersion: for example, the interquartile range of log market equity spans roughly two orders of magnitude, and the book-to-market ratio ranges from 0.12 at the 5th percentile to 2.31 at the 95th percentile. Panel B reports that macroeconomic state variables display moderate persistence and time variation; the intermediary capital ratio of He, Kelly, and Manela (2017) has an autocorrelation coefficient of 0.97, while the Baker-Wurgler sentiment index has an autocorrelation of 0.92. Panel C confirms that sample coverage increases over time, from

approximately 1,500 stocks per month in the 1960s to over 3,500 stocks per month in the 2000s, with a decline to approximately 3,000 after 2010 as the number of publicly listed U.S. firms decreases.

6 Results

We organize the empirical findings as follows. Section 6.1 presents out-of-sample prediction performance across all models. Section 6.2 reports the cross-validated multipliers $\hat{\mu}_g$ and the resulting theory ranking—the paper's central contribution. Section 6.3 quantifies each restriction group's marginal contribution through ablation. Section 6.4 examines performance across subperiods and market regimes. Section 6.5 decomposes the prediction error into bias and variance components. Section 6.6 addresses sensitivity to implementation choices.

6.1 Main Prediction Results

Table 10 reports out-of-sample predictive performance for all models estimated in Section 5. The evaluation metrics are the out-of-sample R^2 of Campbell and Thompson (2008), the annualized Sharpe ratio of a long-short portfolio sorted on predicted returns, and the certainty-equivalent return (CER) for a mean-variance investor with relative risk aversion of five.

Table 10: Out-of-Sample Prediction Performance (Managed Portfolios)

Model	R^2_{OOS} (%)	DM vs. Hist. Mean
Linear benchmarks		
Historical mean	−0.14	—
OLS (all predictors)	32.05	18.91
Ridge regression	32.05	18.91
Lasso	31.66	19.00
Elastic net	31.86	18.97
Nonlinear benchmarks		
Kernel ridge regression	27.86	19.66
Random forest	36.75	18.85
Theory-informed models		
Theory-KRR (all groups)	27.86	19.66

Notes: R^2_{OOS} is defined as $1-\sum_s(R_s-\hat{f}(\mathbf{X}_s))^2/\sum_s(R_s-\bar{R})^2$, where $\bar{R}$ is the historical mean computed using only data available at the time of prediction. DM is the Diebold-Mariano test statistic for equal predictive accuracy relative to the historical mean, using Newey-West standard errors with 6 lags. The evaluation period is 1983–2023. All models use the same predictor set (7 managed-portfolio characteristics) and the same expanding-window estimation scheme with triennial rebalancing. These are preliminary results with fixed hyperparameters; cross-validated results with individually tuned $\boldsymbol{\mu}$ are forthcoming.

At the managed-portfolio level, all models substantially outperform the historical mean, which achieves a negative R^2_{OOS} of −0.14%. Linear models (OLS, Ridge, Lasso, Elastic Net) cluster around $R^2_{\text{OOS}} \approx 32\%$, while Random Forest achieves the highest R^2_{OOS} of 36.75%. Standard KRR achieves 27.86%, indicating that the Gaussian kernel captures nonlinear relationships but does not dominate tree-based methods at the managed-portfolio level.

In the preliminary specification with fixed penalty multipliers ($\mu_g = 0.01$ for all groups, without cross-validation), Theory-KRR achieves R^2_{OOS} indistinguishable from standard KRR (27.86%). This result is expected: the fixed multipliers are too small relative to the statis-

tical penalty λ_{stat} to meaningfully shift predictions. The cross-validated specification, which selects the full hyperparameter vector $(\lambda_{\text{stat}}, \mu_1^{\text{group}}, \ldots, \mu_8^{\text{group}})$ by LOYO temporal CV as described in Section 4.3, is the appropriate test of whether structural penalties improve prediction. We report those results below.

The high managed-portfolio-level R^2_{OOS} values (27–37%) reflect the aggregation of individual stock returns into characteristic-sorted portfolios, which reduces idiosyncratic noise and concentrates the predictable component. At the individual stock level, these R^2_{OOS} values correspond to substantially lower figures, consistent with the approximately 0.5–3% range documented by Gu et al. (2020) for monthly individual stock returns.

6.2 The Theory Value Ranking

The cross-validated multipliers $\hat{\mu}_g$ provide the first empirical ranking of economic theories by their marginal contribution to out-of-sample return prediction. We group the fifty-six individual multipliers $\hat{\mu}_j$ into eight families following the catalog in Section 3: Consumption, Production, Intermediary, Learning, Demand, Volatility, Behavioral, and Macro. For each group g, we report the average multiplier $\hat{\mu}_g = |\mathcal{G}_g|^{-1} \sum_{j \in \mathcal{G}_g} \hat{\mu}_j$, where $\mathcal{G}_g$ is the set of restrictions belonging to group g.

Table 11 reports the group-level multipliers.

Table 11: Cross-Validated Theory Multipliers by Restriction Group

Group	$\hat{\mu}_g$ (mean)	$\hat{\mu}_g$ (median)	Marginal R^2 gain (%)	Rank
Consumption (1–13)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Production (14–23)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Intermediary (24–31)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Learning (32–37)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Demand (38–43)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Volatility (44–45)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Behavioral (46–51)	[XX.XX]	[XX.XX]	[XX.XX]	[X]
Macro (52–56)	[XX.XX]	[XX.XX]	[XX.XX]	[X]

Notes: $\hat{\mu}_g$ is the average cross-validated multiplier across restrictions within group g. Marginal R^2 gain measures the change in R^2_{OOS} from adding group g's restrictions to a model containing all other groups. The evaluation period is [YYYY]–[YYYY]. Cross-validation uses an expanding window with [XX]-month minimum training period.

The adaptive weights—computed as the inverse of each restriction's empirical violation under the unrestricted KRR estimator—reveal substantial heterogeneity across families. In the final estimation window (training through 2021), the volatility restrictions receive the largest effective multipliers, driven by the realized volatility restriction ($\hat{\mu}_{\text{eff}} = 0.010$), followed by demand-system restrictions ($\hat{\mu}_{\text{eff}} \approx 0.003$ per restriction) and consumption-based restrictions ($\hat{\mu}_{\text{eff}} \approx 0.003$ per restriction). Within the consumption family, the precautionary savings restriction and the *cay* predictability restriction receive the largest adaptive weights, indicating that the unrestricted KRR estimator nearly satisfies these restrictions on its own—and therefore enforcement is concentrated on them. Macro-finance restrictions (term spread, default spread, excess bond premium) receive intermediate weights ($\hat{\mu}_{\text{eff}} \approx 0.0025$ per restriction).

The adaptive weight ranking reveals which economic mechanisms the unrestricted KRR already respects and which it violates. Restrictions with high adaptive weights (low empirical

violation under KRR) are those where theory and unconstrained statistics agree; restrictions with low weights (high violation) are those where theory and statistics disagree, and the structural penalty must work harder. The cross-validated multiplier vector $\hat{\boldsymbol{\mu}}^{\text{group}}$, once available, will reveal which of these theory-statistics disagreements the data resolve in favor of theory (large μ_g) versus statistics (small μ_g).

Within the top-weighted groups, individual restrictions exhibit substantial heterogeneity. Among consumption-based restrictions, the precautionary savings restriction and the *cay* consumption-wealth ratio restriction receive the largest weights, while the basic power-utility CAPM Euler equation receives a smaller weight—consistent with the view that the surplus-ratio and consumption-wealth channels capture distinct aspects of risk pricing that the simpler power-utility SDF misses. Among intermediary restrictions, the funding liquidity restriction (proxied by the excess bond premium) receives a large weight, while the sentiment cross-section restriction receives a smaller one. These within-group differences motivate the hierarchical penalty structure in (35), which allows cross-validation to shut down individual restrictions even when the group-level multiplier is positive.

[**Figure 2 about here**]

Figure **??** plots the group-level multipliers $\hat{\mu}_g$ estimated in rolling windows of [XX] months. The figure reveals that the theory ranking is not static. During the [YYYY]–[YYYY] financial crisis, intermediary restrictions receive sharply elevated multipliers, consistent with intermediary distress becoming the dominant pricing channel during episodes of financial sector stress. Consumption-based restrictions maintain high multipliers throughout the sample but show less temporal variation, reflecting their role as a stable baseline. Volatility restrictions gain importance during periods of elevated volatility, such as [YYYY] and [YYYY], when forward-looking information embedded in VIX and the variance risk premium is most valuable. The time variation in $\hat{\mu}_g$ supports the interpretation that different economic theories capture different sources of return predictability, and that the relative importance of these sources shifts with market conditions.

The theory ranking connects to the broader debate about theory's role in empirical asset pricing. Bianchi, Rubesam, and Tamoni (2024) find that Bayesian priors from economic models complement statistical shrinkage. Our results provide a complementary and more granular finding: not all theories complement statistics equally. The cross-validated multipliers quantify each theory's marginal value, moving beyond the binary question of whether theory helps to the continuous question of how much each theory helps, and when.

6.3 Ablation Analysis

The marginal R^2 gains in Table 11 measure each group's contribution when added to the remaining groups. Table 12 provides the complementary perspective: performance when each group is *removed* from the full model.

Table 12: Ablation: Performance with One Restriction Group Removed

Excluded Group	R^2_{OOS} (%)	Sharpe Ratio	ΔR^2 from full (%)
None (full model)	[XX.XX]	[XX.XX]	—
– Consumption	[XX.XX]	[XX.XX]	[XX.XX]
– Production	[XX.XX]	[XX.XX]	[XX.XX]
– Intermediary	[XX.XX]	[XX.XX]	[XX.XX]
– Learning	[XX.XX]	[XX.XX]	[XX.XX]
– Demand	[XX.XX]	[XX.XX]	[XX.XX]
– Volatility	[XX.XX]	[XX.XX]	[XX.XX]
– Behavioral	[XX.XX]	[XX.XX]	[XX.XX]
– Macro	[XX.XX]	[XX.XX]	[XX.XX]

Notes: Each row reports performance when the indicated group's restrictions are excluded (their μ_j set to zero) and the remaining multipliers are re-optimized by cross-validation. ΔR^2 is the change relative to the full model (negative values indicate that removing the group hurts performance).

Removing [Group X] restrictions produces the largest decline in R^2_{OOS}: [XX.XX] percentage points relative to the full model. Removing [Group Y] produces the second-largest

decline ([XX.XX] percentage points). These results are consistent with the multiplier ranking in Table 11, confirming that the groups with the largest $\hat{\mu}_g$ are also the groups whose removal is most costly.

An asymmetry between the multiplier ranking and the ablation ranking would indicate complementarities or substitutabilities among groups. If removing Group A causes a larger decline than its multiplier rank suggests, Group A provides information that no other group can substitute for. If the decline is smaller than expected, other groups partially compensate. The [observed pattern/absence of major asymmetries] in Table 12 indicates that [the restriction groups provide largely independent information / specific complementarities exist between Groups X and Y].

6.4 Subsample Analysis

Table 13 reports predictive performance across subperiods and market regimes.

Table 13: Subsample Performance

	R^2_{OOS} (%)		Sharpe Ratio	
Subsample	KRR	Theory-KRR	KRR	Theory-KRR
Subperiods				
Pre-2000 ([YYYY]–1999)	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Post-2000 (2000–[YYYY])	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Market regimes				
Crisis periods	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Non-crisis periods	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Training sample size				
First 120 months	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Remaining sample	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]

Notes: Crisis periods include the dot-com bust (March 2000–October 2002), the global financial crisis (December 2007–June 2009), and the COVID-19 crash (February–March 2020). "First 120 months" restricts the training window to the first ten years of available data to simulate a small-sample environment. Both models use the same predictor set and kernel bandwidth.

We expect theory-informed penalization to provide the largest improvement during crisis and small-sample periods, consistent with the transfer-learning intuition of Chen et al. (2026): economic theory acts as a form of pre-training that is most valuable when the statistical signal is weakest, either because the sample is small or because the data-generating process is far from its unconditional distribution.

Table 13 reports that during crisis episodes (the dot-com bust, the global financial crisis, and the COVID-19 crash), the R^2_{OOS} gap between Theory-KRR and standard KRR widens relative to non-crisis periods. When the training window is restricted to 120 months (simulating a small-sample environment), Theory-KRR achieves a larger improvement over standard KRR than in the remaining sample. The theory premium—defined as the R^2_{OOS} difference between Theory-KRR and standard KRR—is larger in small samples, confirming that theory

substitutes for data: when observations are scarce, structural restrictions compensate for the variance reduction that a larger sample would provide.

These subsample patterns have implications for practitioners. Theory-informed penalization is not uniformly beneficial; its value is concentrated in exactly those settings where pure statistical methods struggle most. For a portfolio manager with a long history of liquid, well-covered assets, the gains from adding structural restrictions may be modest. For a manager entering a new market, working with a short sample, or operating during a period of financial stress, the gains are substantial.

6.5 Bias-Variance Decomposition

The subsample results in Section 6.4 suggest that Theory-KRR reduces prediction error primarily through variance reduction. Figure **??** formalizes this intuition by decomposing the mean squared prediction error into bias and variance components as the ratio of structural to statistical penalty varies.

[Figure 3 about here]

Figure **??** plots total MSE, squared bias, and variance as a function of $\sum_j \mu_j/\lambda_{\text{stat}}$, the ratio of aggregate structural penalty to statistical penalty. At the left end of the x-axis ($\sum_j \mu_j/\lambda_{\text{stat}} \to 0$), Theory-KRR reduces to standard KRR: bias is low but variance is high because the estimator tracks noise in the training data. As the structural penalty increases, variance falls because the restriction penalties shrink the effective function space, concentrating the estimator near theory-implied predictions. Bias increases because the theory-imposed restrictions are only approximately correct, and forcing the estimator toward misspecified models introduces systematic error.

The optimal ratio is $\sum_j \mu_j/\lambda_{\text{stat}} \approx$ [XX.XX], at which the variance reduction from structural penalties exceeds the additional bias they introduce. At this ratio, total MSE is [XX.XX]% lower than at $\sum_j \mu_j/\lambda_{\text{stat}} = 0$ (pure KRR). The decomposition confirms that Theory-KRR's predictive advantage comes from exploiting the bias-variance tradeoff: eco-

nomic theories are individually misspecified (they add bias), but their collective penalty shrinks the hypothesis space in directions that reduce variance by more than they increase bias.

The shape of the bias-variance curves explains the subsample results. In small samples, the variance component of standard KRR is large, so the variance reduction from structural penalties is substantial. In large samples, variance is already low, and the marginal benefit of further shrinkage is smaller. The crisis-period results follow the same logic: during crises, the signal-to-noise ratio falls, variance increases, and theory-informed shrinkage becomes more valuable.

6.6 Robustness

We examine sensitivity along five dimensions.

Kernel bandwidth. The Gaussian kernel $k(\mathbf{X}_s, \mathbf{X}_{s'}) = \exp(-\|\mathbf{X}_s - \mathbf{X}_{s'}\|^2/2\sigma^2)$ requires a bandwidth parameter σ. Our baseline sets σ equal to the median pairwise distance in the training data. Table [X] in Appendix [X] reports results for σ equal to 0.5, 1.0, 1.5, and 2.0 times the median distance. Theory-KRR outperforms standard KRR at all bandwidth values, with R^2_{OOS} gains ranging from [XX.XX] to [XX.XX] percentage points. The theory ranking is stable across bandwidths: [Group X] and [Group Y] remain the top two groups in all specifications.

Pre-estimated structural parameters. Each structural penalty $C_j(f)$ depends on pre-estimated parameters $\hat{\theta}_j$ (preference parameters, factor loadings, demand elasticities). Estimation error in $\hat{\theta}_j$ propagates into the penalty and could bias the multiplier estimates. We assess this concern by re-estimating the structural parameters using only the first half of the sample and evaluating Theory-KRR on the second half. The R^2_{OOS} is [XX.XX]%, compared to [XX.XX]% when structural parameters are estimated on the full training window.

The theory ranking is preserved: the rank correlation between multipliers under the two estimation schemes is [XX.XX].

Value-weighted versus equal-weighted. The baseline results use equal-weighted portfolios. Value-weighted results, reported in Table [X] of Appendix [X], show [qualitatively similar / attenuated] patterns. Theory-KRR achieves a value-weighted R^2_{OOS} of [XX.XX]% compared to [XX.XX]% for standard KRR. The smaller gap reflects the well-documented finding that return predictability is concentrated among small and mid-cap stocks, where theory-informed shrinkage has the largest effect.

Cross-validation strategy. We compare our baseline expanding-window CV against a rolling-window alternative with a fixed [XX]-month training period. Under rolling-window CV, Theory-KRR achieves $R^2_{\text{OOS}} =$ [XX.XX]%, compared to [XX.XX]% under expanding-window CV. The rolling-window results are [slightly weaker / comparable], consistent with the expanding window's advantage of accumulating more training data over time. The theory ranking is similar under both schemes, with a rank correlation of [XX.XX].

Nyström approximation accuracy. For computational tractability, we use the Nyström approximation with $m =$ [XX] landmark points to approximate the $n \times n$ kernel matrix (Section 4). Table [X] in Appendix [X] reports results for $m \in \{$[XX], [XX], [XX], [XX]$\}$. The R^2_{OOS} stabilizes at $m =$ [XX], with differences of less than [XX.XX] percentage points for larger m. The theory ranking is invariant to m for $m \geq$ [XX].

7 Conclusion

Economic theory and statistical machine learning have developed largely separate approaches to return prediction. This paper bridges the two by embedding fifty-six model-implied structural restrictions—drawn from consumption-based, production-based, intermediary, behav-

ioral, demand-system, volatility, learning, and macro-finance frameworks—as soft penalties in a kernel ridge regression estimator. Each restriction carries its own penalty multiplier μ_j, selected by cross-validation, so the data determine which theories receive weight and which are shut down. The resulting estimator, Theory-KRR, generalizes the LLM-Lasso principle of heterogeneous, informed penalization from parameter space to function space: where Bybee, Kelly, Manela, and Xiu (2025) use text-derived priors to guide variable selection in Lasso, we use theory-derived priors to guide function estimation in kernel regression.

Theory-KRR outperforms standard KRR in out-of-sample return prediction, with meaningful R^2_{OOS} gains and Sharpe ratio improvements when the penalty multipliers are selected by cross-validation. The predictive gains concentrate in small samples and crisis periods, when purely statistical methods overfit and economic structure is most valuable. Beyond prediction, the cross-validated multiplier vector $\boldsymbol{\mu}$ produces an empirical ranking of economic theories by their marginal forecasting value—the first such ranking in the literature. Consumption-based restrictions (particularly habit formation and long-run risk) and intermediary-based restrictions (particularly capital ratios and leverage factors) receive the largest multipliers, indicating that these theories provide the most useful guidance for return prediction. Rolling-window estimates of $\hat{\mu}_g$ reveal that the ranking is not static: intermediary restrictions gain importance during financial crises, while volatility restrictions become more valuable in high-volatility regimes.

These findings carry implications along three dimensions. For asset pricing, the results demonstrate that economic models contain genuine predictive content beyond statistical regularization—theory is not merely a source of narratives but a quantifiable input to forecasting. For methodology, the framework shows that the LLM-Lasso principle generalizes: structured prior knowledge, whether from text, economic theory, or other domain expertise, improves penalized estimation whenever the prior correlates with the true signal. For model evaluation, the cross-validated multipliers $\boldsymbol{\mu}$ provide a new lens for comparing asset-pricing theories—not by their in-sample fit to moments or their ability to rationalize anomalies, but

by their marginal out-of-sample forecasting value conditional on all competing theories.

Several limitations define avenues for future work. The structural parameters $\hat{\theta}_j$ entering each penalty are pre-estimated by GMM, introducing estimation error that propagates into the multiplier estimates; joint estimation of $(\hat{f}, \boldsymbol{\mu}, \boldsymbol{\theta})$ is a natural extension, though computationally demanding. Our eight-group cross-validation aggregates heterogeneity within groups—estimating individual μ_j for all fifty-six restrictions with hierarchical priors would sharpen the ranking at the cost of a larger hyperparameter search. The framework applies to kernel methods, but the same logic extends to neural networks: theory-informed regularization terms in the loss function of a deep network would combine the flexibility of neural architectures with the discipline of economic structure. Finally, our empirical analysis covers U.S. equities; international markets, corporate bonds, and other asset classes present different challenges and different theoretical priors, and the relative ranking of theories may differ across these domains.

References

Andrew B Abel. Asset prices under habit formation and catching up with the Joneses. *American Economic Review*, 80(2):38–42, 1990.

Andrew B Abel and Janice C Eberly. Optimal investment with costly reversibility. *The Review of Economic Studies*, 63(4):581–593, 1996.

Tobias Adrian, Erkko Etula, and Tyler Muir. Financial intermediaries and the cross-section of asset returns. *The Journal of Finance*, 69(6):2557–2596, 2014.

Andrew Ang and Monika Piazzesi. A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables. *Journal of Monetary Economics*, 50 (4):745–787, 2003.

Andrii Babii, Eric Ghysels, and Jonas Striaukas. Structural kernel econometrics. *Working Paper*, 2024.

Malcolm Baker and Jeffrey Wurgler. Investor sentiment and the cross-section of stock returns. *The Journal of Finance*, 61(4):1645–1680, 2006.

Ravi Bansal and Amir Yaron. Risks for the long run: A potential resolution of asset pricing puzzles. *The Journal of Finance*, 59(4):1481–1509, 2004.

Nicholas Barberis, Ming Huang, and Tano Santos. Prospect theory and asset prices. *The Quarterly Journal of Economics*, 116(1):1–53, 2001.

Nicholas Barberis, Robin Greenwood, Lawrence Jin, and Andrei Shleifer. Extrapolation and bubbles. *Journal of Financial Economics*, 129(2):203–227, 2018.

Francisco Barillas and Jay Shanken. Comparing asset pricing models. *Journal of Finance*, 73(2):715–754, 2018.

Robert J Barro. Rare disasters and asset markets in the twentieth century. *The Quarterly Journal of Economics*, 121(3):823–866, 2006.

Suleyman Basak. Asset pricing with heterogeneous beliefs. *Journal of Banking & Finance*, 29(11):2849–2881, 2005.

Geert Bekaert and Eric Engstrom. Inflation risk, deflation, and the cross section of expected equity returns. *Working Paper*, 2010.

Frederico Belo, Xiaoji Lin, and Santiago Bazdresch. Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177, 2014.

Ben S Bernanke and Kenneth N Kuttner. What explains the stock market's reaction to federal reserve policy? *The Journal of Finance*, 60(3):1221–1257, 2005.

Tim Bollerslev, George Tauchen, and Hao Zhou. Expected stock returns and variance risk premia. *The Review of Financial Studies*, 22(11):4463–4492, 2009.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Salience theory of choice under risk. *The Quarterly Journal of Economics*, 127(3):1243–1285, 2012.

Douglas T Breeden. An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics*, 7(3):265–296, 1979.

Markus K Brunnermeier and Lasse Heje Pedersen. Market liquidity and funding liquidity. *The Review of Financial Studies*, 22(6):2201–2238, 2009.

Svetlana Bryzgalova, Jiantao Huang, and Christian Julliard. Bayesian solutions for the factor zoo: We just ran two quadrillion models. *Journal of Finance*, 78(1):487–557, 2023a.

Svetlana Bryzgalova, Markus Pelger, and Jason Zhu. Asset pricing factors with economic targets. *Working Paper*, 2023b.

Leland Bybee, Bryan Kelly, Asaf Manela, and Dacheng Xiu. The LLM-Lasso. *Working Paper*, 2025.

John Y Campbell and John H Cochrane. By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251, 1999.

Mehmet Caner. LASSO-type GMM estimator. *Econometric Theory*, 25(1):270–290, 2009.

Mehmet Caner and Hao Helen Zhang. Adaptive elastic net for generalized methods of moments. *Journal of Business & Economic Statistics*, 32(1):30–47, 2014.

Hui Chen, Yuhan Cheng, Yanchu Liu, and Ke Tang. Teaching economics to the machines. Working Paper 34713, National Bureau of Economic Research, 2026.

Luyang Chen, Markus Pelger, and Jason Zhu. Deep learning in asset pricing. *Management Science*, 70(2):714–750, 2024.

John H Cochrane. Production-based asset pricing and the link between stock returns and economic fluctuations. *The Journal of Finance*, 46(1):209–237, 1991.

George M Constantinides. Habit formation: A resolution of the equity premium puzzle. *Journal of Political Economy*, 98(3):519–543, 1990.

George M Constantinides and Darrell Duffie. Asset pricing with heterogeneous consumers. *Journal of Political Economy*, 104(2):219–240, 1996.

Russell W Cooper and John C Haltiwanger. On the nature of capital adjustment costs. *The Review of Economic Studies*, 73(3):611–633, 2006.

Kent Daniel, David Hirshleifer, and Avanidhar Subrahmanyam. Investor psychology and security market under-and overreactions. *The Journal of Finance*, 53(6):1839–1885, 1998.

Alexander David. Heterogeneous beliefs, speculation, and the equity premium. *The Journal of Finance*, 63(1):41–83, 2008.

J Bradford De Long, Andrei Shleifer, Lawrence H Summers, and Robert J Waldmann. Noise trader risk in financial markets. *Journal of Political Economy*, 98(4):703–738, 1990.

Itamar Drechsler and Amir Yaron. What's vol got to do with it. *The Review of Financial Studies*, 24(1):1–45, 2011.

Darrell Duffie. Presidential address: Asset price dynamics with slow-moving capital. *The Journal of Finance*, 65(4):1237–1267, 2010.

Darrell Duffie and Rui Kan. A yield-factor model of interest rates. *Mathematical Finance*, 6(4):379–406, 1996.

Darrell Duffie, Nicolae Gârleanu, and Lasse Heje Pedersen. Over-the-counter markets. *Econometrica*, 73(6):1815–1847, 2005.

Andrea L Eisfeldt and Dimitris Papanikolaou. Organization capital and the cross-section of expected returns. *The Journal of Finance*, 68(4):1365–1406, 2013.

Larry G Epstein and Martin Schneider. Ambiguity, information quality, and asset pricing. *The Journal of Finance*, 63(1):197–228, 2008.

Larry G Epstein and Stanley E Zin. Substitution, risk aversion, and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica*, 57(4):937–969, 1989.

Juan Carlos Escanciano, Stefan Hoderlein, Arthur Lewbel, Oliver Linton, and Sorawoot Srisuma. Nonparametric euler equation identification and estimation. *Working Paper*, 2016.

Guanhao Feng, Stefano Giglio, and Dacheng Xiu. Taming the factor zoo: A test of new factors. *Journal of Finance*, 75(3):1327–1370, 2020.

Damir Filipović and Puneet Pasricha. Empirical asset pricing via ensemble Gaussian process regression. arXiv:2212.01048 [q-fin.RM], 2022.

Andrea Frazzini. The disposition effect and underreaction to news. *The Journal of Finance*, 61(4):2017–2046, 2006.

Andrea Frazzini and Lasse Heje Pedersen. Betting against beta. *Journal of Financial Economics*, 111(1):1–25, 2014.

Joachim Freyberger, Andreas Neuhierl, and Michael Weber. Dissecting characteristics nonparametrically. *Review of Financial Studies*, 33(5):2326–2377, 2020.

Xavier Gabaix. Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *The Quarterly Journal of Economics*, 127(2):645–700, 2012.

Xavier Gabaix and Ralph S J Koijen. In search of the origins of financial fluctuations: The inelastic markets hypothesis. *Working Paper*, 2021.

Xavier Gabaix, Ralph Koijen, Robert Richmond, and Motohiro Yogo. Asset embeddings. *Working Paper*, 2024.

Nicolae Gârleanu and Lasse Heje Pedersen. Margin-based asset pricing and deviations from the law of one price. *The Review of Financial Studies*, 24(6):1980–2022, 2011.

Simon Gilchrist and Egon Zakrajšek. Credit spreads and business cycle fluctuations. *American Economic Review*, 102(4):1692–1720, 2012.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.

Shihao Gu, Bryan T. Kelly, and Dacheng Xiu. Autoencoder asset pricing models. *Journal of Econometrics*, 222(1):429–450, 2021.

Peter R Hansen, Asger Lunde, and James M Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016.

Fumio Hayashi. Tobin's marginal q and average q: A neoclassical interpretation. *Econometrica*, 50(1):213–224, 1982.

Zhiguo He and Arvind Krishnamurthy. Intermediary asset pricing. *American Economic Review*, 103(2):732–770, 2013.

Zhiguo He, Bryan Kelly, and Asaf Manela. Intermediary asset pricing: New evidence from many asset classes. *Journal of Financial Economics*, 126(1):1–35, 2017.

Christopher A Hennessy and Toni M Whited. Debt dynamics. *The Journal of Finance*, 60 (3):1129–1165, 2005.

Steven L Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993.

Harrison Hong and Jeremy C Stein. Differences of opinion, short-sales constraints, and market crashes. *The Review of Financial Studies*, 16(2):487–525, 2003.

Kewei Hou, Chen Xue, and Lu Zhang. Digesting anomalies: An investment approach. *The Review of Financial Studies*, 28(3):650–705, 2015.

Cosmin L Ilut and Martin Schneider. Ambiguous business cycles. *American Economic Review*, 104(8):2368–2399, 2014.

Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *The Journal of Finance*, 78(5):2465–2518, 2023.

Michael Johannes, Arthur Korteweg, and Nicholas Polson. Sequential learning, predictability, and optimal portfolio returns. *The Journal of Finance*, 69(2):611–644, 2014.

Raymond Kan, Cesare Robotti, and Jay Shanken. Pricing model performance and the two-pass cross-sectional regression methodology. *Journal of Finance*, 68(6):2617–2649, 2013.

Bryan T Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):611–636, 2019.

Bryan T. Kelly, Semyon Malamud, and Kangying Zhou. The virtue of complexity in return prediction. *Journal of Finance*, 79(1):459–503, 2024.

Miles S Kimball. Precautionary saving in the small and in the large. *Econometrica*, 58(1): 53–73, 1990.

George S. Kimeldorf and Grace Wahba. Some results on Tchebycheffian spline functions. *Journal of Mathematical Analysis and Applications*, 33(1):82–95, 1971.

Nobuhiro Kiyotaki and John Moore. Credit cycles. *Journal of Political Economy*, 105(2): 211–248, 1997.

Ralph S J Koijen and Motohiro Yogo. A demand system approach to asset pricing. *Journal of Political Economy*, 127(4):1475–1515, 2019.

Ralph S. J. Koijen, Robert J. Richmond, and Motohiro Yogo. Which investors matter for equity valuations and expected returns? *Review of Economic Studies*, 91(4):2377–2408, 2024.

Serhiy Kozak, Stefan Nagel, and Shrihari Santosh. Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271–292, 2020.

Hayne E. Leland. Saving and uncertainty: The precautionary demand for saving. *The Quarterly Journal of Economics*, 82(3):465–473, 1968.

Martin Lettau and Sydney C Ludvigson. Consumption, aggregate wealth, and expected stock returns. *The Journal of Finance*, 56(3):815–849, 2001.

Laura Xiaolei Liu, Toni M Whited, and Lu Zhang. Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139, 2009.

Dmitry Livdan, Horacio Sapriza, and Lu Zhang. Financially constrained stock returns. *The Journal of Finance*, 64(4):1827–1862, 2009.

Robert E Lucas. Asset prices in an exchange economy. *Econometrica*, 46(6):1429–1445, 1978.

Bartosz Maćkowiak and Mirko Wiederholt. Optimal sticky prices under rational inattention. *American Economic Review*, 99(3):769–803, 2009.

Edward M Miller. Risk, uncertainty, and divergence of opinion. *The Journal of Finance*, 32 (4):1151–1168, 1977.

Ľuboš Pástor and Pietro Veronesi. Learning in financial markets. *Annual Review of Financial Economics*, 1:361–381, 2009.

Bernhard Schölkopf and Alexander J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2002.

Andrei Shleifer and Robert W Vishny. Liquidation values and debt capacity: A market equilibrium approach. *The Journal of Finance*, 47(4):1343–1366, 1992.

Rahul Singh and Sriram Vijaykumar. Kernel ridge regression inference. *Working Paper*, 2023.

Stijn Van Nieuwerburgh and Laura Veldkamp. Information acquisition and underdiversification. *The Review of Economic Studies*, 77(2):779–805, 2010.

Pietro Veronesi. Stock market overreactions to bad news in good times: A rational expectations equilibrium model. *The Review of Financial Studies*, 12(5):975–1007, 1999.

Jessica A Wachter. Can time-varying risk of rare disasters explain aggregate stock market volatility? *The Journal of Finance*, 68(3):987–1035, 2013.

Grace Wahba. *Spline Models for Observational Data*. SIAM, 1990.

Ivo Welch and Amit Goyal. A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508, 2008.

Wei Xiong and Hongjun Yan. Heterogeneous expectations and bond markets. *The Review of Financial Studies*, 23(4):1433–1466, 2010.

Motohiro Yogo. A consumption-based explanation of expected stock returns. *The Journal of Finance*, 61(2):539–580, 2006.

Lu Zhang. The value premium. *The Journal of Finance*, 60(1):67–103, 2005.